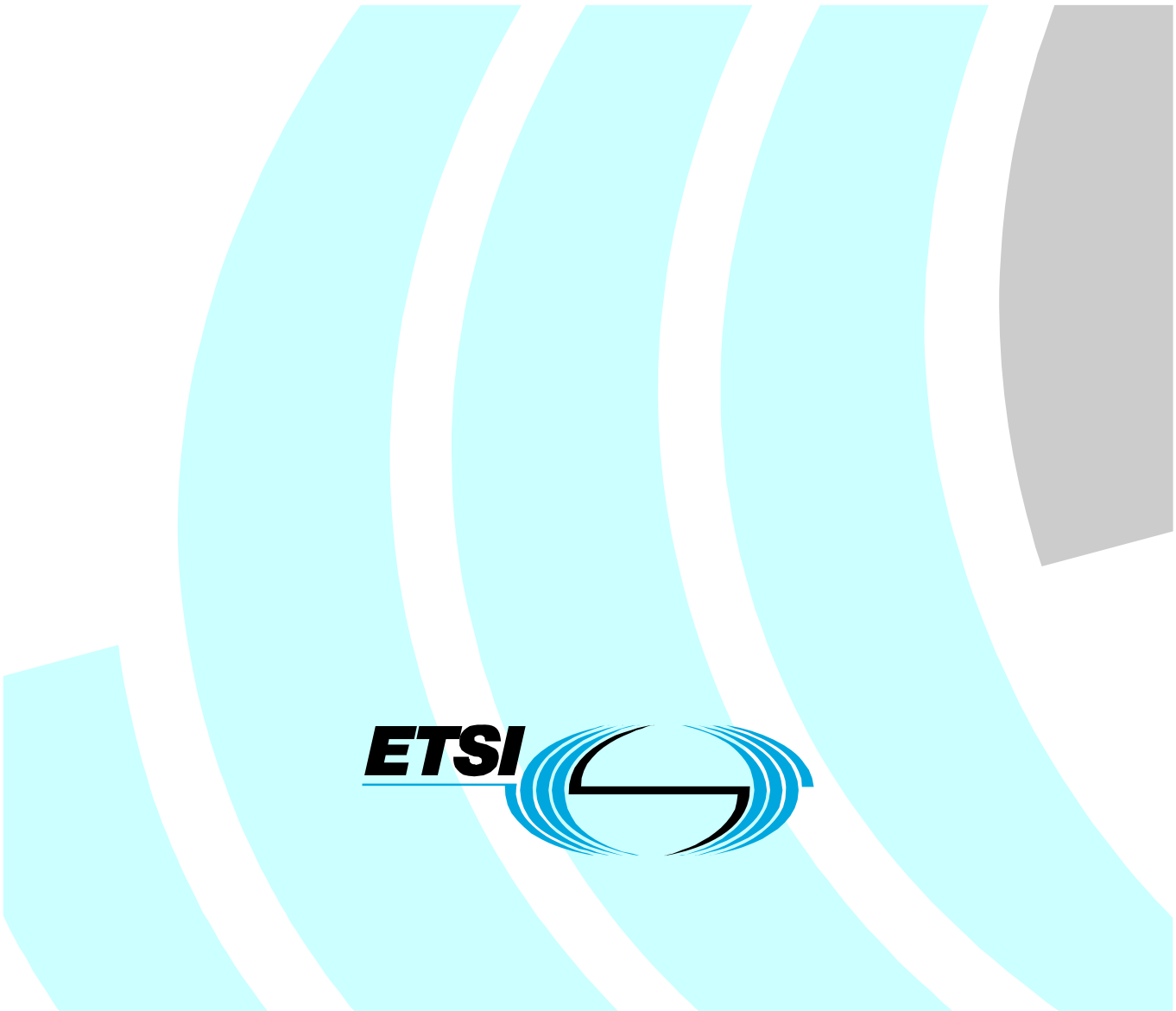
ETSI TR 102 649-2 V1.1.1 (2008-09)

Technical Report

Electromagnetic compatibility and Radio spectrum Matters (ERM); Technical characteristics of Short Range Devices (SRD) and RFID in the UHF Band; System Reference Document for Radio Frequency Identification (RFID) and SRD equipment; Part 2: Additional spectrum requirements for UHF RFID, non-specific SRDs and specific SRDs



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Foreword

This Technical Report (TR) has been produced by ETSI Technical Committee on Electromagnetic compatibility and Radio spectrum Matters (ERM).

The present document includes necessary information to support the co-operation under the MoU between ETSI and the Electronic Communications Committee (ECC) of the European Conference of Postal and Telecommunications Administrations (CEPT).

The present document is part 2 of a multipart deliverable covering RFID and SRD applications in the UHF range.

Part 1: "RFID equipment operating in the range from 865 MHz to 868 MHz";

Part 2: "Additional spectrum requirements for UHF RFID, non-specific SRDs and specific SRDs".

Introduction

The present document requests additional spectrum for UHF RFID, non-specific SRDs and specific SRDs. The additional spectrum is considered necessary because of the rapid growth rate of all these devices and their use in mass market applications. Furthermore the operation of these equipment in globally harmonized frequency bands is highly desirable, especially in view of the forthcoming SRD studies in the ITU-R as a consequence of the Radio Assembly 07 resolution [i.14] and the WRC 07 Resolution [i.15]. This could potentially lead to the global harmonization of spectrum for some of these applications.

1 Scope

The present document applies to UHF RFID, non-specific SRDs such as Home and Building Automation, Telemetry, Data Transmission and specific SRDs such as Metering (water and energy), Alarms, Automotive applications.

The present document describes the development of the RFID and SRDs industries and requests a study into additional frequency designations in the UHF frequency band to meet the medium and long term market requirements of these equipment.

The present document is intended to include all necessary information required by the Electronic Communications Committee (ECC) under the MoU between ETSI and the ECC.

2 References

References are either specific (identified by date of publication and/or edition number or version number) or non-specific.

- For a specific reference, subsequent revisions do not apply.
- Non-specific reference may be made only to a complete document or a part thereof and only in the following cases:
  - if it is accepted that it will be possible to use all future changes of the referenced document for the purposes of the referring document;
  - for informative references.

Referenced documents which are not found to be publicly available in the expected location might be found at http://docbox.etsi.org/Reference.

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NOTE: While any hyperlinks included in this clause were valid at the time of publication ETSI cannot guarantee their long term validity.

2.1 Normative references

The following referenced documents are indispensable for the application of the present document. For dated references, only the edition cited applies. For non-specific references, the latest edition of the referenced document (including any amendments) applies.

Not applicable.

2.2 Informative references

The following referenced documents are not essential to the use of the present document but they assist the user with regard to a particular subject area**.** For non-specific references, the latest version of the referenced document (including any amendments) applies.

[i.1] Commission Decision of 23 November 2006 on harmonization of the radio spectrum for radio frequency identification (RFID) devices operating in the ultra high frequency (UHF) band.

[i.2] "The RFID Revolution: Your voice on the Challenges, Opportunities and Threats".

NOTE: Available at: http://ec.europa.eu/information-society/policy/rfid/doc/rfidswp-en.pdf.

[i.3] BRIDGE Building Radio frequency Identification solutions for the Global Environment, European passive RFID Market Sizing 2007 - 2022, GS1.

[i.4] ERC/REC 70-03: "Relating to the use of short range devices (SRD)".

NOTE: Available at: http://www.erodocdb.dk/doks/implement_doc_adm.aspx?docid=1622.

[i.5] IDTechEx: Boom in RFID will be reflected in Europe's Leading Conference.

NOTE: Available at: http://www.idtechex.com/.

[i.6] Wireless Technology Propels Expansion of European Residential Security Market.

NOTE: Available at: http://www.frost.com/prod/servlet/press-release.pag?Src=RSS&docid=95217860.

[i.7] ETSI EN 302 208 (all parts) (V1.2.1): "Electromagnetic compatibility and Radio spectrum Matters (ERM); Radio Frequency Identification Equipment operating in the band 865 MHz to 868 MHz with power levels up to 2 W".

[i.8] ETSI EN 300 220 (all parts) (V2.1.2): "Electromagnetic compatibility and Radio spectrum Matters (ERM); Short Range Devices (SRD); Radio equipment to be used in the 25 MHz to 1 000 MHz frequency range with power levels ranging up to 500 mW".

[i.9] EC: "Towards an RFID Policy for Europe".

NOTE: Available at: http://ec.europa.eu/information-society/policy/rfid/index-en.htm.

[i.10] ISO/IEC 18000-6 (1st edition; 15 August 2004): "Information technology - Radio frequency identification for item management - Part 6: Parameters for air interface communications at 860 MHz to 960 MHz".

[i.11] ISO/IEC 18000-6 (2004) AMD1 (E) (19 June 2006): "Information Technology - Radio frequency identification for item management - Part 6: Parameters for air interface communications at 860 MHz to 960 MHz, Amendment 1: Extension with Type C and update of Types A and B".

[i.12] CEPT Report 14 (July 2006): "Develop a strategy to improve the effectiveness and flexibility of spectrum availability for Short Range Devices (SRDs) in response to the EU Commission mandate".

[i.13] EPC™ Radio-Frequency Identity Protocols, Class-1 Generation 2 UHF RFID (Version 1.0.9, 31 January 2005): "Protocol for Communications at 860 MHz to 960 MHz".

[i.14] ITU-R Radio Assembly 2007: Resolution 953 Studies to achieve further harmonization for short-range radiocommunication devices (SRDs).

[i.15] ITU-R World Radio Conference 2007 Resolution [COM6/4] (WRC-07) Protection of radiocommunication services from emissions by short-range radio devices.

[i.16] Kyoto Protocol to the United Nations framework convention on climate change (United Nations 1998).

[i.17] EC Directive 2002/91/EC on energy performance of buildings, 16 December 2002.

[i.18] EC Directive 2006/32/EC on energy end-use efficiency and energy services, 5 April 2006.

[i.19] CEPT Report 11: "Strategic plans for the future use of the frequency bands 862-870 MHz and 2400-2483.5 MHz for short range devices".

[i.20] EC Directive 89/106/EEC on the approximation of laws, regulations and administrative provisions of the Member States relating to construction products, 21 December 1988.

[i.21] CEN EN 14604:2005: "Smoke alarm devices".

[i.22] CEPT ECC Report 34: "Compatibility between narrow band digital PMR/PAMR and tactical radio relay in the 900 MHz band, Cavtat, May 2003".

[i.23] CEPT Report 38: "Technical impact of introducing CDMA-PAMR on the UIC DMO & GSM-R radio systems in the 900 MHZ band, Granada, February 2004".

[i.24] CEPT ECC Report 40: "Adjacent band compatibility between CDMA-PAMR mobile services and short range devices below 870 MHz", Granada, February 2004.

[i.25] CEPT ECC Report 41: "Adjacent band compatibility between GSM and CDMA-PAMR at 915 MHz", Granada, February 2004.

[i.26] CEPT ECC Report 58: "Compatibility between TETRA Release 2 TAPS and tactical radio relays in the 870-876 AND 915-921 MHz bands", Stockholm, October 2004.

[i.27] CEPT ECC Report 96: "Compatibility between UMTS 900/1800 and systems operating in adjacent bands", Krakow, March 2007.

[i.28] CEPT ECC Report 37: "Compatibility of planned SRD applications with currently existing radiocommunication applications in the frequency band 863 - 870 MHZ", Granada, February 2004.

[i.29] EN 54-25:2008: "Fire detection and fire alarm systems - Part 25: Components using radio links and system requirements".

[i.30] ETSI-ERM-TG34: 17-07-Analysis-of-RFID-Questionnaire.

[i.31] Commission Decision amending Decision 2006/771/EC on the harmonization of the radio spectrum use by short range devices.

[i.32] ACEA: "Vehicles in use…".

NOTE: Available at: http://www.acea.be/index.php/news/news_detail/vehicles_in_use/.

[i.33] Study on legal, economic & technical aspects of "Collective Use of Spectrum" in the European Community (November 2006) by order of EU Commission.

[i.34] ECC Report 5: "Adjacent band compatibility between GSM and TETRA mobile services at 915 MHz".

[i.35] EN 15232:2007: "Energy performance of buildings - Impact of Building Automation, Controls and Building Management".

[i.36] Directive 98/34/EC of the European Parliament and of the Council of 22 June 1998 laying down a procedure for the provision of information in the field of technical standards and regulations.

[i.37] Directive 98/48/EC of the European Parliament and of the Council of 20 July 1998 amending Directive 98/34/EC laying down a procedure for the provision of information in the field of technical standards and regulations.

3 Definitions, symbols and abbreviations

3.1 Definitions

For the purposes of the present document, the following terms and definitions apply:

alarms: fixed or portable device that uses radio communication for indicating an alert condition at a distant location

channel: small frequency sub-band within the operating frequency band into which a Radio Signal fits

duty cycle: for the purposes of the Recommendation ERC/REC 70-03, the duty cycle is defined as the ratio, expressed as a percentage, of the maximum transmitter "on" time on one carrier frequency, relative to a one hour period

NOTE 1: For frequency agile devices the duty cycle limit applies to the total transmission.

NOTE 2: For specific applications with very low duty cycles and very short periods of transmissions, the definition of duty cycle should be subject to study.

Home and Building Automation: Business and Residential control and system management by radio communication

metering: metering (water and energy) by radio communication

listen before talk: action taken by a device to detect an unoccupied channel prior to transmitting

NOTE: Also known as "listen before transmit".

Short Range Devices (SRDs): radio devices which provide either unidirectional or bi-directional communication and which have low capability of causing interference to other radio equipment

NOTE: SRDs use either integral, dedicated or external antennas and all modes of modulation can be permitted subject to relevant standards. SRDs are normally "license exempt".

specific SRDs: SRDs that are used in specific applications (e.g. Applications of ERC/REC 70-03, annexes 2 to 13)

Tari: reference time interval for a data-0 in Interrogator-to-Tag signalling

NOTE: The mnemonic "Tari" derives from the ISO/IEC 18000-6 (part A) specification [i.10], in which Tari is an abbreviation for Type A Reference Interval.

tag, transponder: device that responds to an interrogation signal

telegram: data transmitted during one duty cycle

tertiary sector (of industry): the service sector, or the service industry

NOTE: This is one of the three main industrial categories of a developed economy.

3.2 Symbols

For the purposes of the present document, the following symbols apply:

E	Electrical field strength
f	frequency
fc	centre frequency
P	Power
d	distance
t	time
λ	Wavelength

3.3 Abbreviations

For the purposes of the present document, the following abbreviations apply:

ACEA	European Automobile Manufacturers' Association

NOTE: See http://www.acea.be/.

AFA	Adaptive Frequency Agility
BRIDGE	Building Radio frequency Identification solutions for the Global Environment
BS	Base Station
CEPT	Conference of Post and Telecommunications Administrations
DC	Duty Cycle
DSSS	Direct Sequence Spread Spectrum
EC	European Commission
ECC	Electronic Communications Committee
e.i.r.p.	equivalent isotropic radiated power
e.r.p.	effective radiated power
ERM	Electromagnetic compatibility and Radio spectrum Matters

EU	European Union
EVVE	The Association for Energy Cost Allocation
FHSS	Frequency Hopping Spread Spectrum
FSK	Frequency Shift Keying
GSM-R	GSM-Railway
HSD	Home Smoke Detector
LBT	Listen Before Talk
MS	Mobile Station
NRE	Non-Recurring Expenditure
OBW	Occupied BandWidth
PAMR	Public Access Mobile Radio
PIR	Passive InfraRed
PLL	Phase Locked Loop
PMR	Private Mobile Radio
RF	Radio Frequency
RFID	Radio Frequency Identification
SME	Small and Medium Enterprises
SRD	Short Range Device
TPMS	Tyre Pressure Monitor Systems
UHF	Ultra High Frequency

4 Comments on the System Reference Document

TC ERM has received a liaison statement from TC TETRA stating that they have no plans to use the dual band 870 MHz to 876 MHz and 915 MHz to 921 MHz. A copy of this letter is attached at annex D.

A number of comments were received from the Nederland's administration and are incorporated in the present document.

5 Executive Summary

The present document requests the designation of additional spectrum in the UHF spectrum for RFID and SRDs. These devices are already installed in large numbers across a wide range of applications within Europe and their use is expected to grow rapidly over the next 10 years. It is anticipated that the current designations of spectrum for RFID and SRDs will be inadequate to meet their future needs. The present document provides independent marketing data that predicts considerable market growth in RFID and SRDs. ECC is requested to undertake a compatibility study in order that additional spectrum may be made available.

The present document identifies two unused frequency bands that could be designated for use by RFID and SRDs. It is believed that for technical reasons these bands cannot be used by the mobile industry (see clause 6.2). It is therefore proposed that the bands are re-designated so that they can be put to productive use.

This proposal provides the following justification for the designation of additional spectrum for RFID and SRDs, as already recognized by the CEPT Report 14 [i.12] (see clause 9.7.3.1) and the Collective Spectrum Use report [i.33], both mandated by the European Commission:

- It is predicted that the use of RFID in Europe will grow dramatically over the next 15 years. As the commercial benefits of RFID become more widely recognized, the technology will be adopted by many new industries. Some of these applications will require improvements to existing RFID performance. Typical examples include greater reading range, improved reading performance, faster data rates and the use of sensors (e.g. temperature, pressure etc) within tags. These requirements can only be met by the provision of additional spectrum. A detailed market analysis is given in annex A.

- The SRD industry has expanded considerably over recent years and has now developed into a number of different industrial sectors. These include metering, automotive applications, alarms, and in wider terms, non-specific SRDs such as home and building automation, telemetry, data transmissions etc. It is anticipated that the present trend in diversification and expansion will continue. An indication of the potential size of the market for SRDs is provided in annex A. Based on these predictions of market growth, it is very evident that additional spectrum will be necessary. This point was already identified in November 2006 in CEPT Report 14 [i.12] in response to a mandate from the EU Commission to develop a strategy to improve the effectiveness and flexibility of spectrum designation for SRDs. The Report recommended that:

  (i) "That CEPT ensures that only the minimum regulations are specified in Recommendation 70-03 and, where appropriate, the application-specific constraints to spectrum use are removed".

  (ii) "New bands should preferably be extensions of SRD bands or close to them".

  (iii) "Introduction of LBT and/or AFA in existing SRD bands is a first priority. However, any benefit from the introduction of LBT and/or AFA may be short lived if the anticipated growth in SRDs occurs. Therefore the identification of new spectrum for SRDs employing these techniques is a second priority".

- The present document has identified two frequency bands, unused by the PMR industry for 15 years. These bands are at 870 MHz to 876 MHz and at 915 MHz to 921 MHz. These could be designated for use by RFID and SRDs. For the reasons given in clause 6.2.1, it is not possible for the mobile industry to operate in these bands. ECC is therefore requested to undertake a compatibility study to determine how these bands could be designated for use by RFID and SRDs.

- As a result of their joint discussions, ETSI_ERM TG28 and TG34 concluded that it would be desirable to separate the high power transmissions of RFID from the low power levels associated with SRDs. The present document therefore proposes that the band 870 MHz to 876 MHz is designated for use by SRDs at less than 100 mW and the band 915 MHz to 921 MHz is designated for high power devices such as RFID (see clause 6.2.2). As important requirement from the industry is that the new SRD bands should be an extension of the present SRD bands or close to them.

- To satisfy the perceived future market requirements for RFID, it is proposed that interrogators will operate in the band 915 MHz to 921 MHz at power levels of up to 4 W e.r.p. in four channels of 400 kHz each. The remainder of the band will be used for the low level response from the tags. This will increase reading performance and potentially permit data rates that are four times faster than those currently possible. Details of the proposed channel plan are provided in annex A.

- Designation of the band 870 MHz to 876 MHz for use by SRDs will satisfy the foreseeable market requirements of the industry. For technical reasons it is proposed to divide the band into two segments. One of these segments covers devices which use duty cycle up to 1 % or LBT with AFA (or equivalent techniques). The other segment is aimed at SRDs that transmit intermittent very short bursts of power and rely on duty cycle for mitigation. The proposed band plan is shown at figure 3. The subdivision of the band into two segments (i.e. non-specific SRDs and specific SRDs) without any further segmentation in each sub-band, fully complies with the spectrum policy for SRDs proposed by the EU Commission. This policy was supported by Member States at the Radio Spectrum Committee meeting of 4-5 December 2006 (see Report 14 [i.12], clause 5.2 for details). This also complies with the key first recommendation of the CEPT Report 14 [i.12] in response to a mandate from the EU Commission to develop a strategy to improve the effectiveness and flexibility of spectrum designation for SRDs (see clause 5.2 of the present document for details). *"That CEPT ensures that only the minimum regulations are specified in Recommendation 70-03 and, where appropriate, the application-specific constraints to spectrum use are removed"*.

5.1 Market Information

5.1.1 RFID

Market information that supports the predicted rapid growth of RFID is provided at clause A.1 of the present document. In addition reference is made to a number of independent market studies such as given in clause A.1.1.

5.1.2 Non-specific SRDs

Market information for non-specific SRDs is provided in clause A.2.

5.1.3 Specific SRDs

Market information for specific SRDs of is provided in clause A.3.

The market information for non-specific SRDs and specific SRDs shows more rapid growth than the conclusions given in 2006 CEPT Report 14 [i.12] called *"Short Range Device Industry: Market and Technology Trends"*. The same applies to similar market studies and investigations contained in the Report on "Collective Spectrum Use" [i.33]. Both Reports are in response to a mandate from the European Commission.

Details are given in clause 7.2.2.2.

5.2 Technical issues

5.2.1 RFID

Clause B.1 shows the channel plan and spectrum mask details for RFID in the proposed band of 915 MHz to 921 MHz.

5.2.2 SRDs

Clauses B.2 and B.3 show the band plan and technical details for non-specific and specific SRDs in the proposed band 870 MHz to 876 MHz. Additional technical information is provided for each of the industrial sectors.

6 Future requirements

The common view of industry, the European Commission [i.33] and CEPT [i.12] is that the 865 MHz to 868 MHz band will be sufficient to cover the immediate spectral needs of UHF RFID. However a lack of spectrum in the UHF frequency range will arise. Unless addressed, this will seriously restrict the forecasted exponential market growth of the industry.

The situation for non-specific and specific SRDs is the same because of their rapidly increasing density and their expansion into a wider range of applications. In particular this is illustrated by the new generation of SRDs in Home and Building automation, metering, alarms and automotive.

A more detailed description of the SRD applications and the justification for additional spectrum is given in clauses A.2 and A.3.

NOTE: The military use of the bands under consideration should be assessed by ECC during consideration of the present document.

6.1 RFID applications

RFIDs are used in item management, logistics and in a wide range of other applications. Details are provided in clause A.1.

Many of these applications require reading ranges of at least 2 meters, and in certain logistics applications ranges from 5 meters to 10 meters, These ranges cannot be provided by alternative technologies and at any other frequency due to the regulatory constraints.

Additional spectrum needs are based on increased RFID usage densities, greater operating distances, and higher data speeds that will permit applications where large numbers of RFID tags are read reliably and quickly.

6.2 SRDs

The proposed band 870 MHz to 876 MHz for use by SRDs will be divided into two homogenous sub-bands (i.e. non-specific SRDs and specific SRDs) without any segmentation in each sub-band.

This fully complies with the recommendations for spectrum policy management in the "Collective Spectrum Use Report" [i.33] as mandated by the EU Commission and supported by Member States at the Radio Spectrum Committee meeting of 4-5 December 2006.

The proposal is based on the specification of a set of common generic characteristics in each sub-band, which ensures compatibility of equipment. This is a step forward compared to the previous approach used in the band 868 MHz to 870 MHz where the band is divided into many application specific sub-bands.

This simplified structure will improve spectrum efficiency, increase opportunities for technical innovation and simplify management of the spectrum.

This proposal also complies with the principal recommendation of CEPT Report 14 [i.12] that:

"CEPT ensures that only the minimum regulations are specified in Recommendation 70-03 and, where appropriate, the application-specific constraints to spectrum use are removed".

A list of applications for non-specific SRDs using either duty cycle or LBT + AFA (or equivalent techniques) is provided in clause A.2. A list of applications for specific SRDs using duty cycle only is provided in clause A.3.

7 Technical Radio Spectrum requirements and justification

7.1 Current regulations

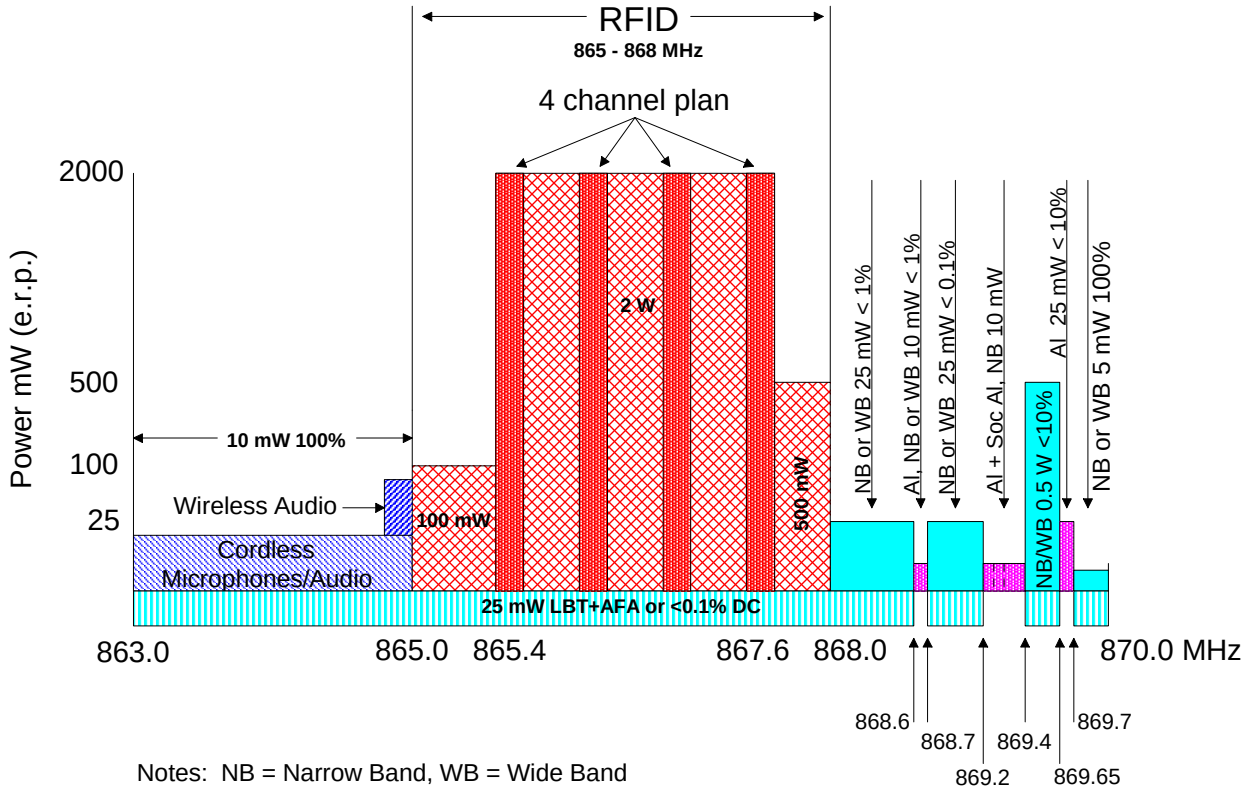
7.1.1 RFID

Operation of RFID in the band 865 MHz to 868 MHz falls under EC Decision 2004/804/EC [i.1] In addition ERC/REC 70-03, annex 11 [i.4] contains a recommendation for RFID in the band 865 MHz to 868 MHz based on 200 kHz reader channels with power levels of up to 2 watts e.r.p., see figure 1. The present document proposes the designation of an additional UHF band to meet the future requirements of RFID.

7.1.2 SRDs

SRDs fall under the EC Decision 2006/771/EC [i.31] However it is anticipated that a revision to this EC Decision will be published shortly.

In addition the present designation of spectrum for SRDs is included in recommendation ERC/REC 70-03, annex 1 [i.4] and is shown below.



Notes: NB = Narrow Band, WB = Wide Band

Band 869.2 - 869.4 sub-divided as follows:-

869.2 - 869.25 Social alarms < 0.1%

869.25 - 869.3 Alarms <0.1%

869.3 - 869.4 Alarms < 1%

Source: CEPT Report 11 [i.19].
For latest and more detailed information consult the latest version of the ERC/REC 70-03 [i.4].

Figure 1: Overview of existing SRD band allocations according to ERC/REC 70-03 [i.4]

Table 1: Excerpt from ERC/REC 70-03, annex 1 [i.4]

	Frequency Band	Power/Magnetic Field	Duty cycle	Channel spacing	ECC/ERC Decision	Notes
f	433.050 - 434.790 MHz	10 mW e.r.p.	< 10 %	No spacing	ECC DEC (04)02	
f1	433.050 - 434.790 MHz	1 mW e.r.p. -13 dBm/10 kHz	up to 100 %	No spacing	ECC DEC (04)02	Power density limited to -13 dBm/10 kHz for wideband modulation with a bandwidth greater than 250 kHz
f2	434.040- 434.790 MHz	10 mW e.r.p.	up to 100 %	Up to 25 kHz	ECC DEC (04)02	
g	863 - 870 MHz (notes 3, 4 and 6)	≤ 25 mW e.r.p.	≤ 0.1 % or LBT (notes 1 and 5)	≤ 100 kHz for 47 or more channels (note 2)		FHSS modulation
		≤ 25 mW e.r.p. (note 6) Power density: -4.5 dBm/100 kHz (note 8)	≤ 0.1 % or LBT (notes 1, 5 and 6)	No spacing		DSSS and other wideband modulation other than FHSS
		≤ 25 mW e.r.p.	≤ 0.1 % or LBT (notes 1 and 5)	≤ 100 kHz, for 1 or more channels (notes 2 and 7)		Narrow/wide-band modulation
g1	868.000 - 868.600 MHz (note 4)	≤ 25 mW e.r.p.	≤ 1 % or LBT (note 1)	No spacing, for 1 or more channels (note 2)	ERC DEC (01)04	Narrow/wide-band modulation No channel spacing, however the whole stated frequency band may be used
g2	868.700 - 869.200 MHz (note 4)	≤ 25 mW e.r.p.	≤ 0.1 % or LBT (note 1)	No spacing, for 1 or more channels (note 2)	ERC DEC (01)04	Narrow/wide-band modulation No channel spacing, however the whole stated frequency band may be used
g3	869.400 - 869.650 MHz (note 4)	≤ 500 mW e.r.p.	≤ 10 % or LBT (note 1)	25 kHz (for 1 or more channels)	ERC DEC (01)04	Narrow/wide-band modulation The whole stated frequency band may be used as 1 channel for high speed data transmission
g4	869.700 - 870.000 MHz (note 4 bis)	≤ 5 mW e.r.p.	up to 100 %	No spacing (for 1 or more channels)	ERC DEC (01)04	Narrow/wide-band modulation No channel spacing, however the whole stated frequency band may be used

NOTE 1: For frequency agile devices the duty cycle limit applies to the total transmission unless LBT is used. For LBT devices without frequency agility, the duty cycle limit applies.
NOTE 2: The preferred channel spacing is 100 kHz allowing for a subdivision into 50 kHz or 25 kHz.
NOTE 3: Sub-bands for alarms are excluded (see ERC/REC 70-03, annex 7).
NOTE 4: The duty cycle, LBT or equivalent technique shall not be user dependent and shall therefore be guaranteed by appropriate technical means.
NOTE 4bis: Audio applications should be excluded. Voice applications allowed with spectrum access technique such as LBT or equivalent technique, the transmitter shall include a power output sensor controlling the transmitter to a maximum transmit period of 1 minute.
NOTE 5: Duty cycle may be increased to 1 % if the band is limited to 865 - 868 MHz.
NOTE 6: For wide-band modulation other than FHSS and DSSS with a bandwidth of 200 kHz to 3 MHz, duty cycle can be increased to 1 % if the band is limited to 865-868 MHz and power to ≤10 mW e.r.p.
NOTE 7: For other narrow-band modulation with a bandwidth of 50 kHz to 200 kHz, the band is limited to 865.5 - 867.5 MHz.
NOTE 8: The power density can be increased to +6.2 dBm/100 kHz and +0.8 dBm/100 kHz, if the band of operation is limited to 865 -868 MHz and 865-870 MHz respectively.

Table 2: Excerpt from ERC/REC 70-03 for Alarms, (annex 7) [i.4]

	Frequency Band	Power		Duty cycle	Channel spacing	ECC/ERC Decs	Notes
a	868.6-868.7 MHz	10 mW	e.r.p.	< 1.0 %	25 kHz	ERC DEC (01)09	The whole frequency band may also be used as 1 channel for high speed data transmissions
b	869.250-869.300 MHz	10 mW	e.r.p.	< 0.1 %	25 kHz	ERC DEC (01)09	
c	869.650-869.700 MHz	25 mW	e.r.p.	< 10 %	25 kHz	ERC DEC (01)09	
d	869.200-869.250 MHz	10 mW	e.r.p.	< 0.1 %	25 kHz	ERC DEC (97)06	Social Alarms
e	869.300 - 869.400 MHz	10 mW	e.r.p.	< 1.0 %	25 kHz	ECC DEC (05)02	

7.2 Proposed Regulation

7.2.1 Justification for proposed new bands

It should be noted that, as of today, no PMR/PAMR installations exist in the European marketplace for the frequency duplex band 870 MHz to 876 MHz paired with 915 MHz to 921 MHz. Neither are there any implementations in these bands of TETRA or CDMA PAMR band class 12. Furthermore a recently approved work item from TETRA shows that they have no future plans to operate in this band (see annex D).

Even when considering the technical feasibility for future projects to operate in this band, it would apply only to a limited number of devices, which would make the business case for PMR/PAMR in this duplex band doubtful, i.e. potential operators will find it difficult to get attractive offers from manufacturers of PMR/PAMR radio equipment for operation in this duplex band. Furthermore, direct mode operation in this duplex band may also raise complex technical co-existence requirements.

The technical reasons behind this notion are considered in the following referenced documents: [i.22], [i.23], [i.24], [i.25], [i.26], [i.27].

The decisive aspect with regard to the feasibility of PMR/PAMR mobile and portable stations (MS) in the duplex band is their receiver selectivity. The adjacent spectrum in Europe is used by GSM/UMTS. (MS to BS in 880 MHz to 915 MHz and BS to MS in 925 MHz to 960 MHz) In this case, an MS receiver still has a separation of 10 MHz from other MS transmissions at 915 MHz. This separation approaches 6 MHz in the case of GSM-R, which is allocated in 876 MHz to 880 MHz (MS to BS) and 921 MHz to 925 MHz (BS to MS). In the GSM-R case the total transmit bandwidth of the band however is limited to 5 MHz, which makes this approach possible. It may be feasible to manufacture PMR/PAMR MS with sufficient Rx selectivity for operation in the upper half of the duplex band but it seems not feasible for operation of the Receiver close to the 915 MHz band edge, i.e. the loss in sensitivity of the receiver is too high. In fact, an investigation of PMR equipment in the market place showed that the lowest separation between uplink and downlink was about 4,5 MHz.

Even if considered technically feasible, the loss in sensitivity of the PMR MS receiver through the intra-system interference would cause a material loss of the cell coverage area. For example consider the following:

1) Assuming wave propagation on the ground, a desensitization of 10 dB in the receiver would cause a reduction of the maximum operating distance between BS and MS of 50 %. Consequently, the loss of covered cell area would be 75 %.

2) If the desensitization increases by 30 dB, the cell radius shrinks to 1/8 and the cell coverage shrinks to 1/64 !

Typically, GSM MS transmissions at or below 915 MHz do not use channel filters. This also applies to the total bandwidth of the uplink. In the GSM case, GSM MS handles this aspect by never transmitting and receiving at the same time (and not by filtering).

This fact has been known since 1992 when it was discussed in CEPT. Later on, ECC Report 5 [i.34] only investigated the scenario of TETRA TAPS BS interfering with a GSM BS. The scenario GSM MS into PMR/PAMR MS was not considered.

In some European countries, frequency managers have received a dispensation from the military to use the band also for civilian purposes.

7.2.2 Proposals and justification for RFID, non-specific SRDs and specific SRDs

A summary of the main requirements covering RFID, non-specific SRDs and specific SRDs is provided below:

a) it is proposed to extend the current regulation for band (g) of annex 1 of ERC/REC 70-03 [i.4] from 863 MHz to 870 MHz to also include the 870 MHz to 873 MHz (see note of figure 3) frequency range for narrow and wide band use with a 1 % D.C or LBT + AFA or equivalent mitigation technique (see table 3);

b) it is proposed to extend the current regulation for band (g) of annex 1 of ERC/REC 70-03 [i.4] from 863 MHz to 870 MHz to the 863 to 873 MHz (see note of figure 3) frequency range for FHSS and DSSS systems with a 0,1 % D.C or LBT + AFA or equivalent mitigation technique (see table 3);

c) it is proposed to designate the frequency range of 873 MHz to 876 MHz (see note of figure 3) for devices with low duty cycles that can not co-exist in the same frequency range with current LBT + AFA devices (see table 3).

d) it is proposed to designate spectrum for UHF RFID high performance interrogators as shown in table 4 within the frequency range from 915 MHz to 921 MHz;

e) it is proposed that SRDs should share the band 915 MHz to 921 MHz with RFID. This will provide additional spectrum for SRDs and will be of particular benefit in situations where higher powers are required. To ensure compatibility between equipment, SRDs should use either LBT + AFA or Duty Cycle or an equivalent technique.

This proposal is in accordance with the recommendations of the EC (including RSCOM) [i.33] and of CEPT [i.12]. These recommendations promote the co-existence of multiple types of equipment within bands by the use of common technical characteristics.

Table 3: Proposal for SRDs

Frequency bands	Power	Duty cycle	Channel bandwidth	Notes
g4 869,7 MHz to 870 MHz (see note 4bis of table 1)	≤ 25 mW e.r.p.	1 % D.C. or LBT + AFA	No spacing (for 1 or more channels)	Narrow/wide-band modulation No channel spacing, however the whole stated frequency band may be used. Audio applications excluded. Voice applications allowed with LBT together with 1 minute carrier time-out timer (see note)
	≤ 5 mW e.r.p.	Up to 100 % D.C. or LBT + AFA		
g5 870 MHz to 873 MHz (see note of figure 3) non-specific SRDs	≤ 25 mW e.r.p.	1 % D.C. or LBT + AFA	no spacing	Narrow/wide-band, DSSS with 0,1 % duty cycle permitted FHSS duty cycle t.b.d. and spectrum density vs. minimum channels to be studied
g6 873 MHz to 876MHz specific SRDs Short burst telegrams (see note of figure 3)	≤ 1 mW e.r.p. (to be studied) ≤ 25 mW e.r.p. ≤ 100 mW e.r.p.	Up to 5 % D.C Up to 1 % D.C Up to 0,1 % D.C (see figure 2)	No channel spacing	Narrow/wide-band, DSSS with 0,1 % duty cycle permitted FHSS duty cycle and t_{on} time of hops to be studied [i.28]
NOTE: Taking advantage of the present spectrum re-farming plans for SRDs, it is also proposed to include devices with up to 25 mW e.r.p. and up to 1 % D.C. or LBT + AFA in the band 869,7 MHz to 870 MHz. For the power and duty cycle values of the frequency range g6, the trade-off of varying power and duty cycle can be interpolated from figure 2.				

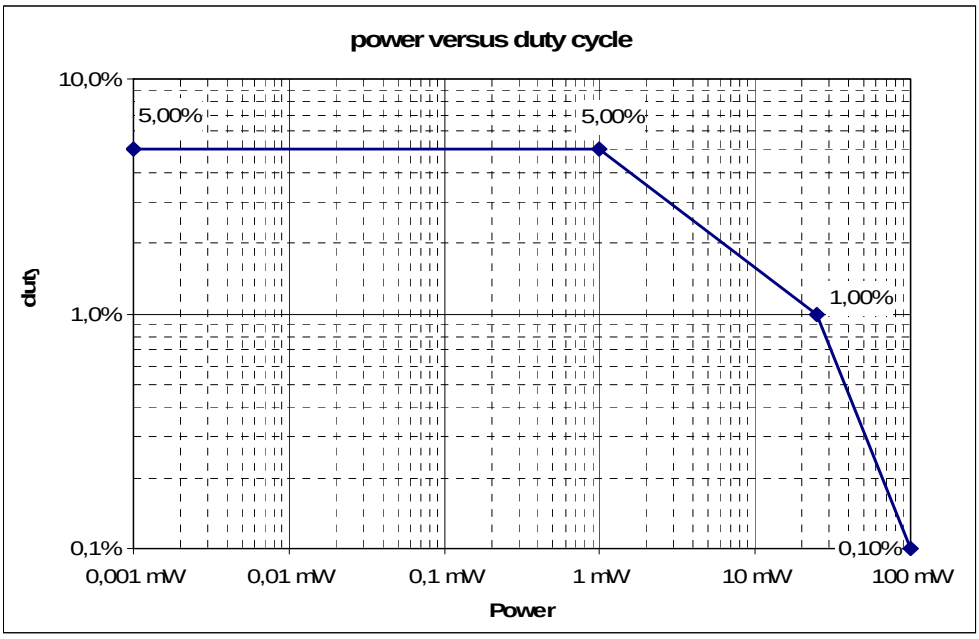
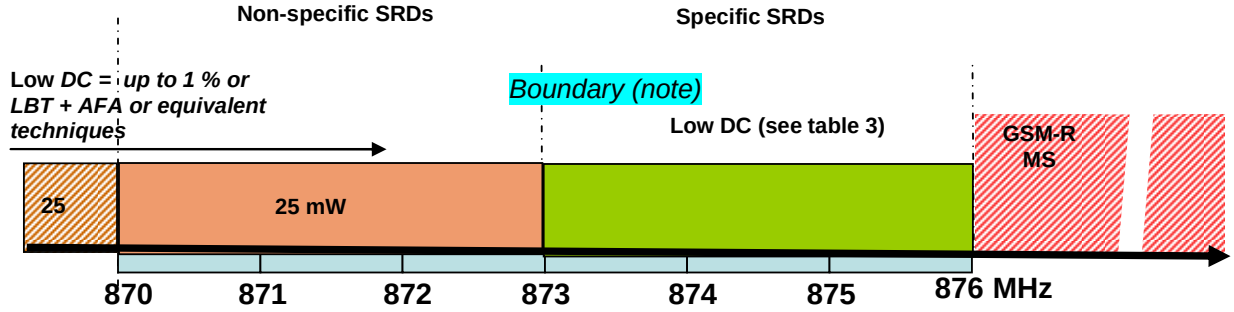


Figure 2: Allowed $P_{e.r.p.}$ versus Duty cycle in the 873 MHz to 876 MHz band (see note of figure 3)

Figure 3 displays the proposed frequency allocation for the band 870 MHz to 876 MHz for non-specific and specific SRDs.



NOTE: It is expected that the boundary between non-specific and specific SRDs will lie somewhere between 873 MHz and 874 MHz. The exact frequency will be determined following the compatibility study when the impact of GSM-R on the upper part of the band has been quantified. It should be noted that specific SRDs have a minimum requirement for 2 MHz of usable spectrum.

Figure 3: Proposal for 870 MHz to 876 MHz frequency range

Table 4: Proposal for high performance RFID interrogators and SRDs

Frequency bands	Power	Duty cycle	Channel bandwidth	Notes
Interrogators: 915 MHz to 921 MHz Interrogator centre frequencies 916,2 MHz 917,4 MHz 918,6 MHz 919,8 MHz	≤ 4 W e.r.p. on a single interrogator channel for each individual interrogator	No mandatory limit for transmitter on-time. However interrogators will not be allowed to transmit longer than it is necessary to perform the intended operation	f_c ± 200 kHz	Interrogators may operate in any of the four high power channels
Tags: Between 915 MHz to 921 MHz	< -10 dBm e.r.p. per tag		f_c ± 1 000 kHz for tag response	
SRDs: 915 MHz to 921 MHz Center frequencies for high power SRD channels 916,2 MHz 917,4 MHz 918,6 MHz 919,8 MHz	≤ 0,1 W e.r.p. in RFID high power channels	0,1 % duty cycle or LBT + AFA	f_c ± 200 kHz	Transmit levels outside of high power channels will not be allowed to exceed 25 mW e.r.p.
NOTE: fc are the carrier frequencies of the interrogators. SRD receivers should be category 2 or better as specified in EN 300 220 [i.8]. To minimize the risk of interference from RFID, SRDs may use LBT with AFA or equivalent techniques in the high power channels. Suitable separation distances should be studied. To minimize the risk of interference from SRDs to RFID tag responses, SRDs should use LBT with AFA or equivalent techniques in the remaining 2,2 MHz. Suitable separation distances should be studied.				

Figure 4 is a graphical representation of the table 4 proposal for high performance RFID interrogators and SRDs.

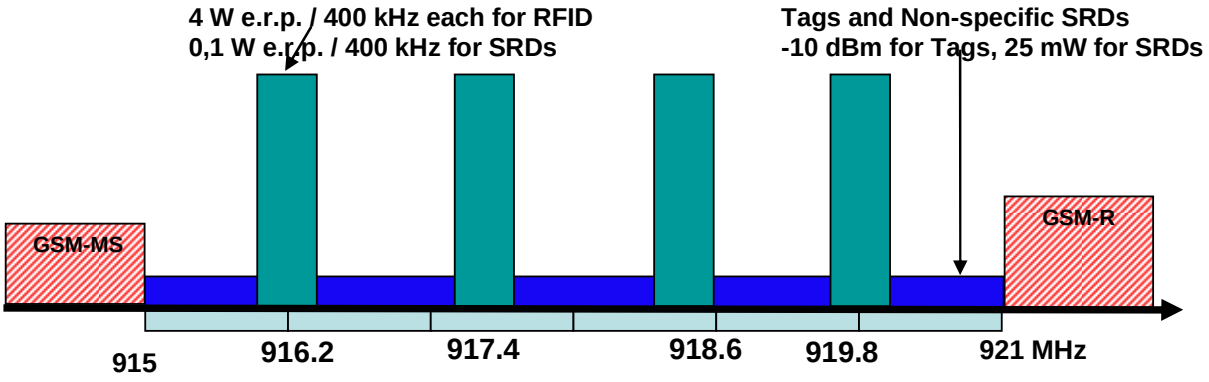


Figure 4: Proposal for high performance RFID and SRD applications

It should be noted that the separation between channels of 1,2 MHz permits the higher data rates required for the new applications and ensures that the receive bands do not overlap into the adjacent high power channels.

7.2.2.1 Justification for spectrum requirements for RFID

Market studies, as well as answers received from questionnaires, have demonstrated that additional spectrum will be required to accommodate a growing installed base and to offer increased performance. The high performance systems will meet the needs for higher data rates and longer reading ranges. Higher data rates will be used either for reading large numbers of tags or for future applications involving the movement of tagged items at greater speeds.

The main conclusions from the marketing information are:

- RFID at UHF will grow exponentially over the next 10 to 15 years.
- Increased usage densities, higher data speeds, faster reading of higher numbers of tags, improved reading performance all require additional spectrum for operations at higher power levels. This should preferably be in the range around 915 MHz since this is the same frequency band used for RFID by many of the major non-European trading partners.
- Medium and long term spectrum planning according to the present document is needed in order to provide the regulatory environment that will ensure the continued growth of RFID at UHF and enable the market predictions to be realized.

7.2.2.2 Justification for spectrum requirements for non-specific SRDs

During 2005 - 2006 market studies were conducted by the major European industry associations i.e. EICTA and LPRA (see more in clause A.2). These studies included the detailed analysis of replies to questionnaires from association members. The results from this work provided a valuable contribution towards the CEPT Report 14 [i.12]. The present document was prepared in response to a mandate from the EU Commission to develop a strategy to improve the effectiveness and flexibility of spectrum designation for SRDs.

Similar market studies and investigations were conducted by a consortia of consultants (i.e. Mott MacDonald Ltd, Aegis System Ltd and Indipen) who also produced a Report on "Collective Use of Spectrum" [i.33] in response to a mandate from the EU Commission.

Both Reports demonstrated that additional spectrum will be required to accommodate the rapid growth in SRDs. In that regard a specific recommendation in both Reports advised that *"New bands should preferably be extensions of SRD bands or close to them"*.

Furthermore the CEPT Report 14 [i.12] stated that "Introduction of LBT and/or AFA in existing SRD bands is a first priority. However, any benefit from the introduction of LBT and/or AFA may be short lived if the anticipated growth in SRDs occurs. Therefore the identification of new spectrum for SRDs employing these techniques is a second priority".

A generation of SRDs using LBT + AFA has already been sold in large volumes with favorable results. For example, based on the experience of several millions of installed devices using LBT and AFA, the Home and Building automation industry reports that the technique is performing very satisfactorily.

Already the sale of devices in the Home and Building automation sector exceeds estimates in the CEPT Report 14 [i.12] by 25 % (see clause A.1.2 for market details).

Information from the other non-specific SRD applications (see examples below) indicates that they will also continue to grow. For example telemetry is already producing, an annual turnover of approximately € 1 000 million. The combined effect of Building and Home Automation together with these other applications will place increasing pressure on the existing band 863 MHz to 870 MHz leading in the medium term to an unacceptable deterioration in performance.

7.2.2.3 Justification for spectrum requirements for specific SRDs (e.g. Metering of water and energy) and alarms

The requirements for the metering market are defined by the EC directive 2002/91/EC [i.17] and 2006/32/EC [i.18]. Both of these Directives focus on the efficient use of energy to mitigate anthropogenic climate change and to reduce the economic dependency of the EC on the import of primary energy resources. Metering and individual cost allocation are known to reduce significantly the consumption of primary energy resource by stimulating a change in consumers' behaviour.

Furthermore, it has been shown that metering and control systems with radio technology (according to class B systems EN 15232 [i.35]) will reduce total energy consumption by about 10 %.

The new requirements for time resolution for statistical data and control cannot be met without radio interfaces.

Figure 5 shows a "specific" SRD deployment in the home environment.

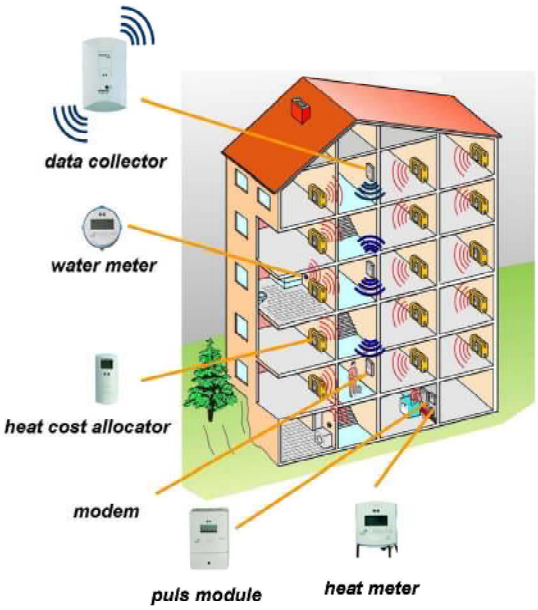


Figure 5: Scenario for specific SRDs

Due to the need for flexibility of use and low cost of installation, alarms also include radio devices and are battery powered. This holds not only for smoke detectors, but also for any kind of intrusion detection sensor, such as a PIR (passive infrared) sensor for motion detection.

In order to be compliant with European harmonized standards, developed under the mandates M/109 and M/139 of the European legislation Construction Products Directive 89/106/EEC [i.20], a minimum service life of 3 years (battery) is necessary for fire detection systems. It is also a legal requirement to provide a response time of less than 10 sec and detection of defective devices within a few minutes.

Alarms also include personal alarms. See clause A.3.1 for market details.

7.2.2.4 Justification Automotive applications

Short Range Devices perform a variety of important functions in modern automobiles. In accordance with information provided by the European Automotive Manufacturers Association (ACEA), the existing passenger car fleet in Europe (ACEA: "Vehicles in use… ": in 2005) [i.32] consists of more than 250 million vehicles. Currently approximately 60 % of these vehicles are equipped with one or more SRDs. Approximately 6 million new vehicles are sold in Europe every year. 80 % of all new vehicles are currently equipped with SRD devices. This percentage is increasing and will reach 100 % in the near future.

In addition the variety of different SRDs used in vehicles is also increasing. This includes traditional remote keyless entry systems, which are developed further into passive entry systems and personal car communication systems. Also safety related systems such as Tyre Pressure Monitoring Systems (TPMS) and Truck-trailer communication systems are increasing as well as deployment of security systems including vehicle alarm systems.

TPMS is considered to be an integral part of measures required by the European Commission in order achieve the EU policy targets for CO_2 emissions.

A considerable number of malfunctions of the keyless entry systems are reported from customers and this trend will further increase as a consequence of the increase in general SRD applications operating in the existing frequency bands.

Access to new spectrum within the band 873 MHz to 876 MHz (see note of figure 3) provides the potential for a long term solution for automotive Short Range Devices.

8 Main conclusions

In order to achieve the commercial needs of the market it is important that spectrum is made available as soon as possible. The ECC is therefore requested as a matter of priority to consider the designation of additional spectrum for use by RFID and SRDs in the UHF frequency range.

8.1 UHF RFID

An up-to-date and comprehensive market study is given in [i.3] on the future size of the RFID industry. This study projects exponential growth for UHF RFID applications.

The need for high usage densities, higher data speeds, greater ranges, faster reading of increased numbers of tags, all require additional spectrum in a timely manner.

To satisfy the above requirements it will be necessary to designate more spectrum that will provide high performance reader channels.

An additional frequency designation in the range 915 MHz to 921 MHz is proposed for high performance RFID systems. This will harmonize the operation of RFID in Europe with many of the other major non-European countries as shown in figure A.1.

8.2 SRDs

Market studies have shown that the demand for SRDs is set to rise dramatically in response to end-user requirements.

Existing spectrum available for SRDs is insufficient to satisfy this growing demand, so an additional frequency designation in the range 870 MHz to 876 MHz as shown in figure 3 is requested.

CEPT Report 14 [i.12] and the "Collective Spectrum Use" Report [i.33], which were both prepared in response to a mandate from the EU Commission, have noted that additional spectrum will be required to accommodate the rapid growth of SRDs. A specific recommendation in both Reports advised that: *"New bands should preferably be extensions of SRD bands or close to them"*.

Furthermore the CEPT Report 14 [i.12] stated that *"Introduction of LBT and/or AFA in existing SRD bands is a first priority. However, any benefit from the introduction of LBT and/or AFA may be short lived if the anticipated growth in SRDs occurs. Therefore the identification of new spectrum for SRDs employing these techniques is a second priority"*.

Already the sale of devices in the Home and Building automation sector is exceeding estimates in the CEPT Report 14 [i.12] by 25 % (see clause A.2 for market details). This rapid growth is expected to continue. Similarly information from the other non-specific SRD applications indicates that they will also continue to grow.

The rapidly emerging market for specific SRDs with their own particular technical requirements cannot be accommodated within the existing 868 MHz to 870 MHz band.

The combined effect of non-specific and specific SRDs will place increasing pressure on the existing band 863 MHz to 870 MHz leading in the medium term to an unacceptable level of performance.

9 Requested ECC and EC actions

ETSI requests ECC to consider the present document, which includes the necessary information to support the co-operation under the MoU between ETSI and the Electronic Communications Committee (ECC) of the European Conference of Post and Telecommunications Administrations (CEPT).

ECC is requested to undertake studies on the proposals for new spectrum for high performance UHF RFID systems, for non-specific SRDs and for specific SRDs such as automotive, metering and alarm applications. It is requested that these studies are performed within a time frame of 12 months.

It is expected that the ECC and EC actions will result in harmonized European conditions for the availability and use of the radio spectrum for RFID and SRDs.

It should be noted that in some European countries the military have frequency assignments in the bands being proposed for SRD and RFID applications. As part of the requested action, the ECC is therefore invited to coordinate the civil use of the bands with the military. It is requested that ERC/REC 70-03 [i.4] be amended to include designated spectrum for the new frequency ranges and applications.

10 Expected ETSI actions

ETSI ERM TG34 intends to create a revision of the existing EN 302 208 [i.7] in line with the proposed changes and ECC study results.

ETSI ERM_TG28 intends to create a revision of the generic EN 300 220 [i.8] in line with the proposed changes and ECC study results. In addition, it is intended to generate a new harmonized standard for the specific SRDs as outlined in the present document.

Annex A:
Detailed market information - Market size, Applications and requirements

A.1 RFID

A.1.1 Market evolution

Market forecasts indicate a very high growth rate for RFID applications. Some market studies are noted as follows:

1) The "*Final report from CEPT in response to the second mandate to CEPT to develop a strategy to improve the effectiveness and flexibility of spectrum availability for SRDs*", Chapter 6 entitled: "*Short Range Device Industry: Market and Technology Trends*" [i.12].

2) The Bridge study "European passive RFID market sizing 2007 - 2022" [i.3].

3) IDTechEx: Evolution of RFID, 1,71 billion tags sold in 2007, 2006-2017 from $2,77 billion to $27,88 billion [i.5].

4) RFID questionnaire to end users as well to the RFID industry. See clause A.3.

5) Excerpt from EC "Towards an RFID Policy for Europe":

   "*….cumulative sales of RFID tags have totalled 2.4 billion over the past 60 years, with 600 million tags being sold in 2005 alone, the value of the market, including hardware, systems and services, is expected to be multiplied by 10 between 2006 and 2016. The number of tags delivered in 10 years will be over 450 times the number actually to be delivered this year*" [i.9].

The European Commission is funding the BRIDGE project (Building Radio frequency Identification solutions for the Global Environment) [i.3].The project will develop easy-to-use technological solutions for the European business community including SMEs. This will ensure a basis for collaborative systems for efficient, effective and secure supply chains.

Based on the market analysis carried out under the BRIDGE project, it is predicted that in five years more than 170 000 RFID readers will be deployed in Europe at 30 000 locations. During this period these readers will process a total of about 3 billion tags. These numbers will grow significantly and by 2022, it is expected that more than 6 million readers will be operating at 450 000 locations, with about 86 billion tags purchased annually.

It is believed that these numbers are conservative, as they only represent a small percentage of the total potential number of objects that can be tagged. For example, the forecast is based on the estimate that in 2012 approximately 2 % of all items in retail will be tagged. In 2022 the forecast is that roughly 25 % of all non-food items and 5 % of all food items in retail will be tagged. If we experience a technology breakthrough in the next fifteen years that reduces the cost of an RFID tag to less than one cent, these numbers could increase dramatically. In particular the number of tags on food items could grow to hundreds of billions.

The European Commission has published the results of a consultation "The RFID Revolution" [i.2] stating that the present designation of spectrum for RFID at UHF is considered sufficient for the initial deployment of RFIDs but will be inadequate once the technology becomes ubiquitous.

Another publication by the Commission notes: "*Current trends and forecasts indicate that the RFID market will grow fast in the next 10 years. While cumulative sales of RFID tags have totalled 2.4 billion over the past 60 years, with 600 million tags being sold in 2005 alone, the value of the market, including hardware, systems and services, is expected to increase by a factor of 10 between 2006 and 2016. The number of tags delivered in 10 years will be over 450 times the number actually to be delivered in 2007*" [i.9].

The shortage of spectrum for UHF RFID in the medium and long term will arise due to:

- The proliferation of the RFID technology in industry, manufacturing, logistics and many applications in daily life.
- The need for higher data transmission speeds which requires larger bandwidth. This is necessary for reliable reading of data and also for writing data to tags.
- The penetration of "lossy" objects or packaging materials in stacks or pallets requires higher power and/or multiple reading at higher data transmission speeds providing redundancy in order to enhance reading performance. This applies especially in applications with fast moving goods like conveyor belts or production lines.

New RFID frequency designations will be needed to provide additional capacity for present systems. In addition they should offer improved performance to meet the needs of more demanding applications. Preferably they should operate at around 915 MHz where RFID tags are optimized for global markets. A possible European frequency band outside the cluster of bands around 915 MHz would suffer a loss in performance (see figure A.1).

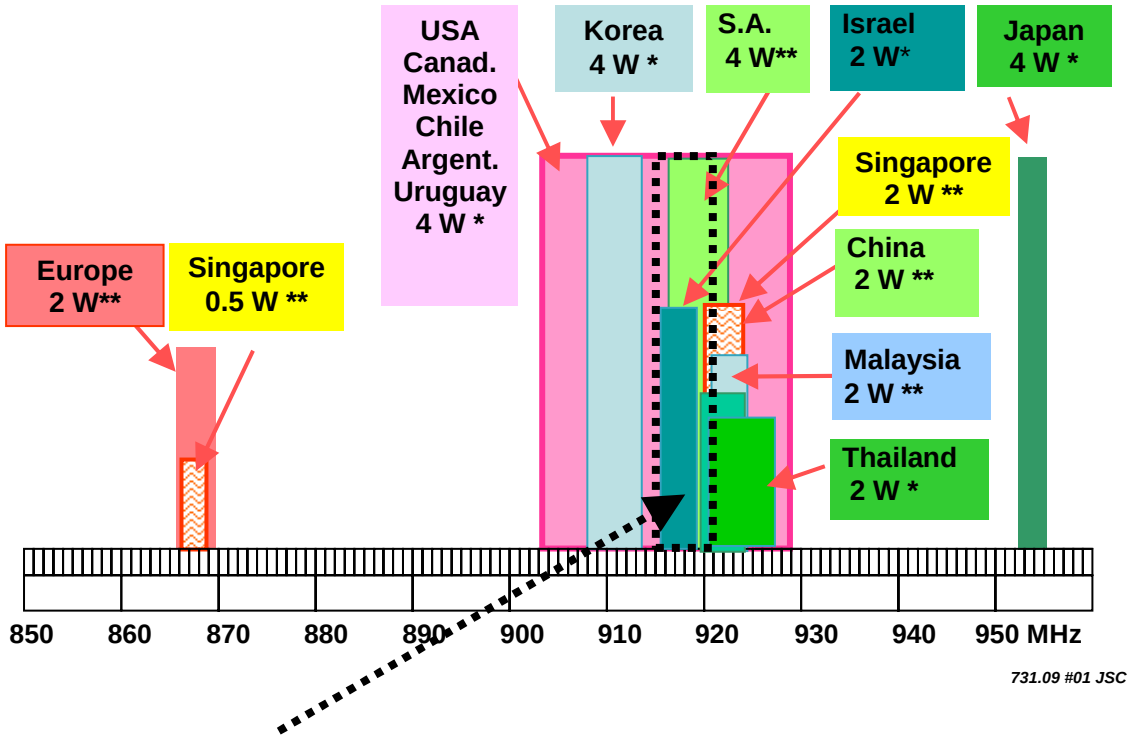
When considering a new frequency designation for RFID, the following criteria have to be considered:

- The designated spectrum for UHF RFID in many other countries outside of Europe is considerably larger than in Europe. Also UHF RFID in Europe operates in a lower frequency band than used by other major trading countries.
- A reader bandwidth of more than 200 kHz is needed in applications using higher data speeds or reading larger numbers of tags. Typical examples of such applications include rapidly moving conveyor systems and pallets carrying large numbers of tagged items.
- To compete with global performance in reading reliability, range, object penetration and also to be competitive in the global environment, sufficient spectrum for RFID should preferably be designated within the range of 900 MHz to 930 MHz. If not European industry will suffer a degradation in range, reading reliability and speed as compared to the countries which have allocations around 915 MHz.

A list of potential applications for RFID is provided below:

- automatic article identification;
- asset tracking;
- airline baggage handling;
- security and alarm systems;
- waste management;
- proximity sensors, anti-theft systems;
- location systems;
- data transfer to handheld devices;
- automotive and general manufacturing automation;
- wireless control systems;
- transport and logistics;
- military;
- libraries;
- medical and hospital applications;
- postal services;
- e-government.

A diagram showing the present designation of spectrum for RFID in Europe in relation to the frequencies used for RFID by the other major trading countries is given below in figure A.1. It will be seen that the frequencies used in Europe are separated by almost 50 MHz from the band that has most generally been adopted by the other regions.



Legend for power levels: e.i.r.p. *, e.r.p.: **

Figure A.1: Proposal for harmonization of RFID in Europe with the other Regions

Given the growing global deployment of RFID, there is no doubt that performance will continue to improve and tag prices will come down leading to yet further adoption of the technology. It seems highly likely that, in the future, RFID at UHF will become an integral part of business in Europe and other major industrialized countries. This will lead to an installed base of millions of interrogators reading billions of tags each year. Furthermore Europe is embedded in the global trading, and manufacturing environment. It is thus important that Europe conforms as far as possible to global trends.

Requirements

According to a recent survey [i.9], the logistics' industry and RFID system providers have stated that:

- The density of RFID tags in a typical load or transport unit can be up to 1 000 to 1 500 tagged objects. The separation between tags typically can be between 2 mm - 3 mm to 30 cm.
- According to the application the minimum separation between fixed interrogators as well as hand/mobile readers can be between 0 meter to 1 meter. This means that multiple channels are required in order to avoid tag to tag cross-interference.
- The reading range should be up to 9 meters.
- The time available for reading all tags in a bulk/pallet environment varies from 0,5 sec to 2 sec max. Also the data carried by a tag can vary from 96 bits to 256 bits.
- Acceptable reading performance on a pallet should approach 100 %. This requires higher power as well as high data rates for increased read redundancy.
- Additional features (e.g. monitoring of temperature, shock, humidity, pressure,) will be used in critical applications to assure the condition of delivered goods. This may require higher power and much increased data capacity.
- The performance of European RFID systems and tags should match global performance. Presently this cannot be guaranteed since European frequencies and power levels do not provide comparable reading performance to e.g. US, SA, Far East, Canada and South American countries.

A.1.2 Questionnaire on RFID

A Questionnaire was circulated by TG34 to system integrators and end users in order to obtain supplementary information not included in the marketing studies. In particular it was necessary to know the densities of RFID interrogators that are likely to exist on busy sites and the extent to which these interrogators will be fixed or portable devices. The report on the survey is available as input document 17_07 at the ERM_TG34 docbox and is called "Analysis of replies to Questionnaire on RFID" [i.30].

The principal conclusions from analysis of the responses to the Questionnaire are summarized below:

1) The responses were unanimous that RFID would lead to a reduction in costs and improved levels of efficiency within their business. These benefits will flow through to customers.

2) All of the respondents focussed their attention on the use of RFID for logistics and materials control. However it seems probable that over the next 15 years a much more varied range of applications will emerge.

3) Consideration should be given to channel planning in order to ensure that the predicted high densities of interrogators can co-exist satisfactorily.

4) The maximum reading ranges of up to 15 meters requested by end users will not be achievable with passive tags. This would require transmit levels in excess of 10 W e.r.p., which would be unacceptable to regulators and would also raise concerns over health and safety. It active tags are used at these ranges, it will be necessary to review their maximum permitted output levels.

5) A few end users requested the ability to read at rates of up to 2 000 tags per second. If a link frequency of fc ±1 000 kHz is selected this will permit a reading rate in excess of 800 tags per second for tags holding 96 bits of data. However in practice some redundancy in the reading process is necessary so this figure will probably not be achieved in practice. The reading rates will thus be lower than those requested by some end users.

6) It seems very probable that in future tags will be introduced that will include sensors.

A.2 Non-specific SRDs

A.2.1 Market evolution

There is a strong market demand for non-specific SRDs in almost all areas of daily life including the enhancement of safety measures and caring e.g. for handicapped and elderly people.

As stated in clause 7.2.2.2 SRD market studies were conducted during 2005 to 2006 by the major European industry associations i.e. EICTA and LPRA. These studies included the detailed analysis of replies to questionnaires from association members. EICTA also sent questionnaires to all EU National Industry Associations. The results from this work provided a major contribution towards the section in CEPT Report 14 [i.12] called *"Short Range Device Industry: Market and Technology Trends"*.

CEPT Report 14 [i.12] was prepared in response to a mandate from the EU Commission to develop a strategy to improve the effectiveness and flexibility of spectrum designation for SRDs.

Similar market studies and investigations were conducted by a consortia of consultant firms (i.e. Mott MacDonald Ltd, Aegis System Ltd and Indipen) who also produced a Report on "Collective Spectrum Use" [i.33] in response to a mandate from the EU Commission.

Both Reports have demonstrated that additional spectrum will be required to accommodate the rapid growth in SRDs. In that regard a specific recommendation in both Reports advised:

"New bands should preferably be extensions of SRD bands or close to them".

Furthermore the CEPT Report 14 [i.12] stated:

"Introduction of LBT and/or AFA in existing SRD bands is a first priority. However, any benefit from the introduction of LBT and/or AFA may be short lived if the anticipated growth in SRDs occurs. Therefore the identification of new spectrum for SRDs employing these techniques is a second priority".

Several million SRDs have been installed by the Home and Building automation industry using LBT and AFA. The industry reports that this technique is performing very satisfactorily.

Already the sale of devices in the Home and Building automation sector exceeds estimates in the CEPT Report 14 [i.12] by 25 % (see clause A.1.2 for market details).

Other non-specific SRD applications will continue to grow at a steady rate from a significant base. For example telemetry is producing an annual turnover of approximately € 1 000 million. The combined effect of Building and Home Automation together with these other applications will place increasing pressure on the existing band 863 MHz to 870 MHz leading in the medium term to an unacceptable deterioration in performance.

A non-exhaustive list of applications for SRDs using either duty cycle or LBT + AFA (or equivalent techniques) is provided below.

- Home and Building automation (some examples):
  - Lighting control
  - Shutter control
  - Awnings and blinds control
  - Windows, doors and gates openers control, garage doors
  - Electrical door lock systems
  - Heating regulation
  - Air conditions control
  - Swimming pool surveillance and control
  - Ventilation
  - Combined scenarios
  - Presence Simulation
  - Automatic controls for comfort energy saving and security purposes
  - Sensors (temperature, wind, light, rain)
  - Presence monitoring
- Telemetry and telecommand (some examples):
  - Pumping station monitoring
  - Electricity network monitoring
  - Crane and machinery control
  - Oil industry installations
- Mixed speech and data (some examples):
  - Wireless door entry
  - Alarm ambiance background scanning
  - Baby and elderly monitoring
- Access control (some examples):
  - Disabled persons access
  - Security applications
- Machine to Machine (some examples):
  - Remote data collection (state of machines)
  - Remote control (management)
  - Real time telemetry of sensors
  - Remote payment
  - Remote restaurant/bar customer orders data collection
  - Portable Bar Code Scanner

- Aviation and Maritime applications (some examples):
  - Remote data maintenance collection (service information of aircraft downloaded while taxiing)

A.2.2 Home and Building automation

The European Home and Building automation market includes applications such as shutters, terrace awnings, blinds and curtains, electrical door locks, electrical windows, garage door and gate openers, heating control, lighting control, etc…

For the last decade, the above applications have mainly been used as stand-alone devices, where users controlled only a single device at a time - *e.g. up, down, stop for a shutter or a opening/closing a door*. Simple command protocols were used.

Now the need for more **comfort**, more **security** and a strong emphasis on **energy saving** require the use of advanced system control protocols. These protocols permit the use of dynamic functions that coordinate the operation of actuators though advanced automatic controls and sensors - *e.g. a feature that is currently in high demand is the "indoor climate control" where the air inside the house remains healthy and fresh with a minimum of energy loss. When the humidity exceeds the desired level, specified roof and vertical windows are opened for a short period.*

The wide range of applications falling under Home and Building automation represents the main market area for non-specific SRDs. Initially many devices, such as shutters and garage doors were mechanical. Today at least 50 % of shutters and 70 % of garage doors are controlled remotely using SRDs. With the arrival of low cost chip sets the industry has largely migrated to the use of LBT + AFA due to the superior performance of this technology and its ability to operate in congested environments.

By 2003/2004 the Home and Building automation industry had already installed large volumes of first generation RF products based on unidirectional communication and Duty Cycle techniques. These first generation products provided simple and convenient control of electrical blinds, shutters, doors, etc.

Sales of Home and Building Automation devices in 2005 and 2010 are outlined in the CEPT Report 14 [i.12].

Table A.1: Sales of Home and Building Automation

Parameters	2005	2010
SRD units supplied annually	5 million	10 million
Systems installed to date	10 million	70 million
Annual Turnover	€ 10 000 million	€ 15 000 million
Work force	1 million	2 million

As the market matures volumes will rise and functionality will increase. France, Spain, Italy and Benelux now represent mass markets where customers purchase sophisticated Home and Building automation systems, In these countries home automation systems already exceed 50 % of electrical equipment in buildings. In 2006 the European SRD industry installed around 15 million Home and Building automation devices and a further 20 million alarm systems. The growing density of these devices means that co-existence will become of increasing importance.

During 2007 Home and Building Automation manufacturers sold more than 17 million devices, **which exceeded by 25 % the prediction in CEPT Report 14 [**i.12**].** This growth was achieved as a consequence of considerable investment by manufacturers. Market data shows that this trend will continue. Therefore the number of units installed by 2010 will exceed by 25 % the forecast of 70 million devices in CEPT Report 14 [i.12].

Nowadays the need for more **comfort**, more **security** and a strong emphasis on **energy saving** requires the use of sophisticated system control protocols. These protocols permit the use of dynamic functions that coordinate the operation of actuators though advanced automatic controls and sensors - *e.g. a feature that is currently in high demand is the "indoor climate control" where the air inside the house remains healthy and fresh with a minimum of energy loss. When the humidity exceeds the desired level, specified roof and vertical windows are opened for a short period.*

When a Home and Building automation system is installed in a residential house, it typically comprises fifty nodes (15 shutters, 2 garage door opener, 1 gate opener, 2 electrical door locks, 2 terrace awnings, 4 roof windows, 2-5 electrical vertical windows (ventilation control), 1 intrusion alarm, 8 heating zones and a dozen lighting terminals) This represents a sophisticated control system requiring advanced two-way communications (see figure A.2).

Such a system allows a user, who is leaving the home, to close down the system by means of a single command. This command might for example lock the door, set the alarm, close the shutters and reduce the heating level.

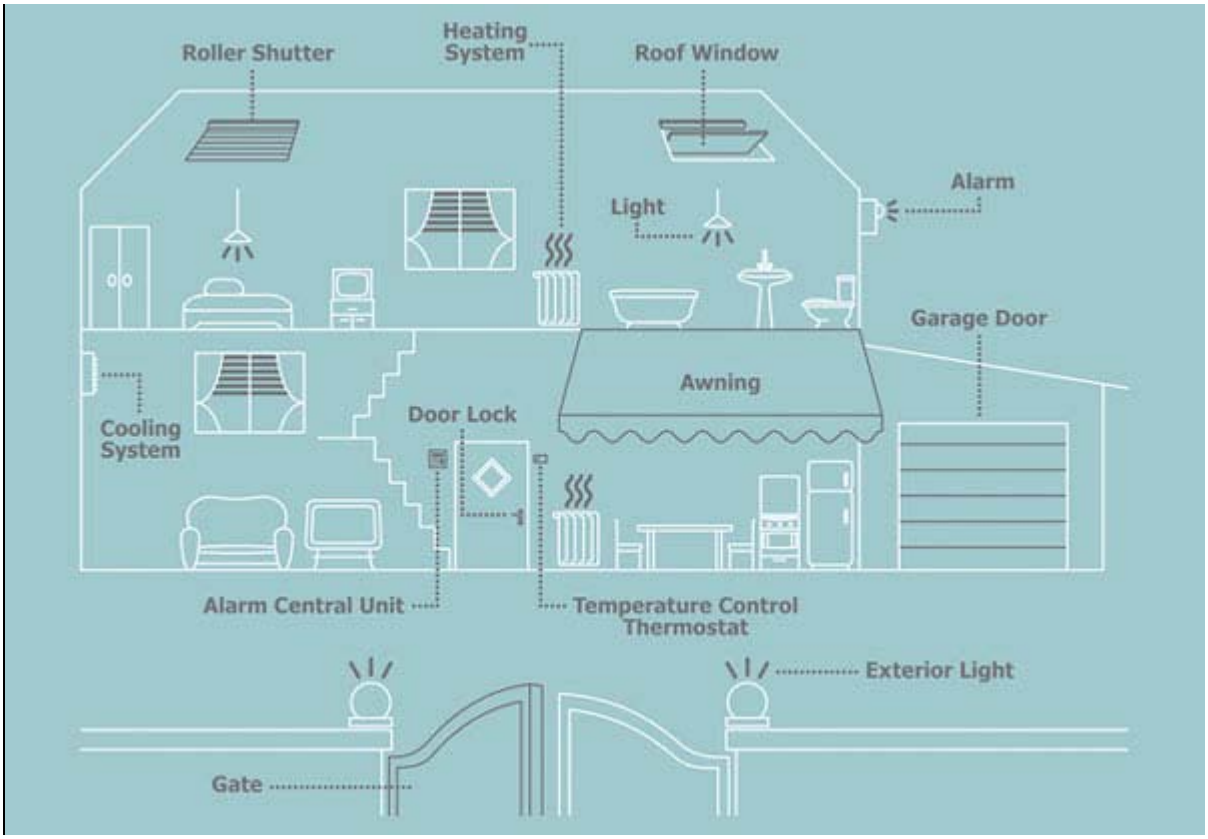


Figure A.2: Home and Building Automation applications



Figure A.3: Examples for Home and Building Automation

With growing interest in energy saving, home automation systems will place increasing emphasis on natural environmental control. This will include more use of automatic ventilation, solar panels and other similar techniques. Lighting will also form part of the home automation system since it can be used to provide energy savings and improved security.

The latest version of EN 300 220 [i.8] recognizes the continued need for low-end, low-cost products while also specifying advanced techniques for use in next generation products.

Technological facts

Home and Building automation applications require:

1) **Short latency:** a rapid response to transmitted commands is essential.

2) **Signal transmission reliability:** the control address should be guaranteed to reach the target.

3) **Point-to-point communications** should be able to cover a whole building environment without recourse to the use of repeaters.

4) Control devices are mainly **battery powered** (or similar) with 5 to10 years lifetimes to virtually eliminate the need for regular maintenance (inside walls and ceilings).

5) A **medium data rate** in the range of a few tens of kilobits per second (kbps) e.g. 19,6 kbps to 38,4 kbps, that represents a balanced compromise between acceptable current consumption, radio range and functionality.

6) **No duty cycle restriction** to allow a dialogue between the controllers (wall switches, remote control, IP gateway, etc.), the actuators (windows, heating boiler, lighting, etc.) and the sensors. While a usage pattern cannot be predicted, typically it would be low (< 1 %).

Technologies above 1 GHz are unsuitable due to the poor propagation inside buildings. An operating frequency in the 863 MHz to 870 MHz and 433 MHz bands is therefore necessary.

Up to 2003/2004 home automation applications were mainly designed with one-way communication and a low duty-cycle (433,92 MHz). This provided only a limited set of controls for the application.

These old systems are being replaced by a new generation of bi-directional systems which offer increased benefits. Operation in the band 870 MHz to 873 MHz using LBT and AFA brings the following benefits:

- The ability to operate on any one of a number of channels provides greater reliability and low latency.
- When combined with stringent transmission time control, LBT with AFA provides an efficient mitigation technique that allows coexistence between multiple systems.

Home and Building automation systems are designed with an efficient link budget using at least category 2 receivers. Typically a transmitter power of less than 25 mW e.r.p. is adequate, requiring a receiver sensitivity is in the range of -105 dBm or better.

This allows the end-user direct point-to-point control of all actuators within his premises from his roof window to his garage door in the basement. This is necessary since customers are unwilling to pay the added cost for repeaters.

A.2.3 Various non-specific SRDs

Other applications of non-specific SRDs include:

- Telemetry and telecommand.
- Mixed speech and data.
- Access control.
- Machine to machine.
- Aviation and maritime.

Some examples of applications for these other non-specific SRD sectors are given below.

Many of the above non-specific SRD sectors have adopted LBT with AFA due to the operational benefits and the increased availability of low cost integrated circuits.

The descriptions and views expressed in CEPT Report 14 [i.12] are still largely valid.

Non-specific SRD applications in the band 915 MHz to 921 MHz are also proposed subject to a positive outcome from the spectrum sharing study. The major potential benefit for SRDs would be the ability to operate at higher powers than are possible in the bands 863 MHz to 870 MHz and 870 MHz to 873 MHz (see note of figure 3). This may be beneficial for example for some telemetry applications that would be prepared to share the spectrum using polite techniques such as LBT + AFA (i.e. if a channel is occupied by RFID, the SRD will move to an unoccupied high power channel). From a spectrum policy point of view, this approach is consistent with the EC (RSCOM) and CEPT objectives, which seek to broaden the types of equipment that may operate within a single band. This contrasts with the present rigid sub-bands within the band 868 MHz to 870 MHz, which are each devoted to a specific application. A market trend similar to Home & Building automation applies to other non-specific SRD applications. Major examples include:

- Telemetry and telecommand (some applications):
  - Pumping station monitoring
  - Electricity network monitoring
  - Crane and machinery control
  - Oil industry installations

- Mixed speech and data (some applications):
  - Wireless door entry
  - Alarm ambiance background scanning
  - Baby and elderly monitoring
- Access control (some applications):
  - Disabled persons access
  - Security applications
- Machine to Machine (some applications)
  - Remote data collection (state of machines)
  - Remote control (management)
  - Real time telemetry of sensors
  - Remote payment
  - Remote restaurant/bar customer orders data collection
  - Portable Bar Code Scanner
- Aviation and Maritime applications (some applications):
  - Remote data maintenance collection (service information of aircraft downloaded while taxiing)

Summarizing:

Congestion in 868 MHz to 870 MHz band will increase with the growth not only of Home and Building Automation but also of the above sectors (For example, telemetry is producing an annual turnover of approximately € 1 000 million).

The conclusions reached in the CEPT Report 14 [i.12] on the above non-specific SRD applications are still largely valid.

Many of these applications have already adopted frequency agility and LBT techniques in the 868 MHz to 870 MHz spectrum due to the rapid availability of components at low cost.

A.3 Specific SRDs

The predicted growth rates for SRD applications in the field of metering and alarms will lead to a mass market for these devices. Prominent applications include Residential Metering Alarms and Energy Control.

There is a requirement for approximately 3 MHz of spectrum (see figure 3) in the specific SRD band to be designated for use by Residential Metering, Alarms and Energy Control. This will be shared with Automotive using only duty-cycle mitigation techniques.

It has been determined that these applications can co-exist within the same spectrum, given the inherent geographical separation between their normal areas of operation as well as the very low duty cycles involved.

The following is a non-exhaustive list of applications that covers Metering, Alarms and Automotive applications and includes:

- Alarms:
  - portable personal security alarms
  - intruder alarm systems
  - smoke warning and fire detection device
  - change of state alert
  - leakage detection alert

- Metering:
  - heat and water cost allocation
  - heat consumption metering and usage monitoring
  - metering device monitoring
  - energy and water control applications
  - hot and cold water metering
  - leakage detection
  - gas, oil and electricity metering and usage monitoring
  - continuous smoke detector status monitoring

A.3.1 The range of Metering and Alarms Applications

Currently consumption based billing is applied in more than 25 million dwellings. According to an analysis by the Association for Energy Cost Allocation (EVVE), the total market size is about 49 million Dwellings. The number of meters used for consumption based billing varies from region to region depending on climatic conditions, the services offered, building's standard etc. At the moment about 125 million Meters are used for heat and water cost allocation and this number is expected to double within the next 10 years.

Battery operation is a requirement due to the metering points being on radiators, water and heating pipes and ceilings (for HSDs). For economic use of the equipment, the market requires a service life of at least 10 years.

For most devices, the capacity of the battery is below 1,3 Ah. Less than half of the capacity is available for the radio interface. This means that average currents for the radio circuit are below 7 µA.

To minimize power consumption the maximum duty cycle of the transmitters in the meters is less than 0,1 %. However data collectors have to communicate with multiple devices and therefore should operate at duty cycles of between 0,1 % and 1 %.

To circumvent the high internal resistance of the small long-life batteries, buffer capacitors are used to supply the peak currents necessary during transmission. The size of the capacitors and the required transmit power limits the transmit time. This means that the transmit time should be very short, which requires a data rate in the range of 100 kbit/s. Taking into consideration the environmental and design constraints, this leads to channel bandwidth requirements of 250 kHz (see clause B.2).

The electronic compartment of a typical metering device is less than 15 cm long and the area is less than 80 cm². This space should accommodate the printed circuit board, battery and antenna. Both the metering circuit and the radio interface have to fit on the same printed circuit board. The limited space means that the device is not suited to frequencies below UHF.

As the metering devices will be distributed over the whole building, transmission through multiple walls is required in order to communicate with the data collectors. Due to the limited power available and the significant increase in attenuation with frequency, the required range cannot be achieved above 1 GHz. So, significantly higher frequencies are not suitable.

For efficient use of metering equipment, the distance between metering devices and data collectors is in the range of 15 meters to 25 meters, which corresponds to the distance across a typical residential building. A typical propagation path will pass through 3 to 5 walls. It has been found in practice, that 10 to 25 mW e.r.p. provides adequate propagation without overshooting into neighbouring dwellings.

Today 12 % of the installed meters are equipped with a radio interface. Due to changing requirements the market share for meters with radio technology is rapidly growing. The long service life of 5 to 10 years will ensure the continued long term growth of meters with radio interfaces. Members of the EVVE expect a market share of 66 % by 2017.



Figure A.4: Wireless detector devices

With the installation of growing numbers of smoke detectors, the density of monitoring devices in buildings will progressively increase. To simplify their monitoring and maintenance, it is probable that many of them will be interlinked by radio. To meet the requirements of the EC Legislation Directive 89/106/EEC for construction products [i.20] together with new regional building regulations, two harmonized standards (EN 14604:2005 [i.21] and the new EN 54-25:2008 [i.29] have been published. These standards have triggered interest amongst alarm service providers in offering home smoke detectors. Members of the E.V.V.E. expect to have more than 24 million Smoke alarms with radio technology in the market in the next 10 years.

A market study issued by Frost & Sullivan in 2006 [i.6] identified one of the key factors driving the European security market as follows:

"Wireless technology - The many advantages of wireless technology makes it one of the key drivers propelling the European residential security market. In 2005, wireless equipment made up for more than 50 percent of the total security intrusion alarms installation. Wireless is especially popular in retrofit and stimulates replacement".

This study included not only intrusion alarms, but also fire detection and smoke warning devices used for residential applications.

A.3.1.1 Metering and Alarms Details

Already there are more than 15 million devices installed with a market potential of a further 165 million devices. One of the most important applications is telemetric measurement of water and energy consumption by energy service providers.

The commitment to the Kyoto protocol in 1998 [i.16] has resulted in several measures being taken by the EU to reduce CO_2 emissions. According to the European Parliament, 40 % of the total consumption of primary energy is caused by buildings and the tertiary sector [i.17]. That is why energy savings in buildings will contribute significantly to the overall energy efficiency. The EC Directive on energy performance of buildings says:

"Increased energy efficiency constitutes an important part of the package of policies and measures needed to comply with the Kyoto Protocol and should appear in any policy package to meet further commitments" [i.17].

The EC directive 2006/32/EC on energy services [i.18] requires member states to ensure the provision of statistical information about the energy consumption of the end user. This means measurement of energy and water consumption using metering at many different points within individual dwellings. It is known that metered billing of services encourages users to reduce their energy consumption.

Residential metering, also known as sub-metering, cannot be provided with the required time resolution without telemetry. Therefore radio communication is becoming an important feature in the metering business. Currently more than 15 % of the market is equipped with radio interfaces and the market is growing rapidly. Sub-metering is already provided in about 25 million dwellings, equivalent to 50 % of the potential market in the EC.

For most devices, the capacity of the battery is below 1,3 Ah. Less than half of the capacity is available for the radio interface. This means that average currents for the radio circuit are below 7 µA.

To minimize power consumption the maximum duty cycle of the transmitters in the alarms is less than 0,1 %. However communications gateways have to communicate with multiple devices and therefore should operate at duty cycles of between 0,1 % and 1 %.

In addition, to circumvent the high internal resistance of the small long-life batteries, buffer capacitors are used to supply the peak currents necessary during transmission. The size of the capacitors and the required transmit power limits the transmit time.

In order to achieve maximum flexibility over acceptable distances, wireless alarm systems may use multi-hop or meshed networks. A drawback of such systems is the reduced response time per hop, which makes it difficult to comply with the regulations. This problem may be overcome by the use of higher data rates, which inevitably means more bandwidth.

It is not possible to operate alarm systems acceptably in situations in which they share the band with other devices that may transmit for extended periods (say up to 2 sec). In such circumstances alarm systems would have to use much higher duty cycles in order to meet the required response time. This would reduce their battery life to unacceptable levels. Alarm systems should therefore operate in bands that only permit the use of equipment with short duty cycles and low transmitting power. Due to the quantity of data transmitted, and since the energy consumption of a device is proportional to the transmission time, data rates in the range of 100 kbit/s are needed. Taking into consideration the environmental and design constraints, this leads to channel bandwidth requirements of 250 kHz.

The dimensions of the housing of a typical fixed installed alarm device is less than 100 mm in diameter and height, which should contain the detector part including sounder, radio module, battery and antenna. This tight restriction on space means that the device is not suited to operate at frequencies below UHF because of the size of the antenna. The required range cannot be achieved at frequencies above 1 GHz due to the limited transmit power of the device and the significant increase in attenuation through walls. So, higher frequencies are also not suitable.



Figure A.5: Wireless smoke detector

Currently regional building regulations in Europe require the installation of home smoke detectors to reduce the number of people killed and injured in fires.

Modern commercial home smoke detectors offer a radio interface to interlink the various detectors within the dwelling to provide enhanced protection for vital escape routes. Long term reliability needs adequate monitoring and maintenance. Metering service providers are ideally placed to offer this service, which can effectively be provided by using the same radio system as used by metering.

For efficient use of the spectrum, the interface for both metering and alarm equipment, should preferably be compatible and use the same radio band. A total provision, installation and maintenance package can then be offered by metering service providers, using the same infrastructure as for energy metering and cost allocation.

At present, all metering applications use the frequency range 863 MHz to 870 MHz, as identified in ERC/REC 70-03 [i.4] for non-specific SRD devices. Two factors have changed the situation:

- The sub-bands identified for non-specific SRDs now have LBT + AFA sharing criteria, whereas previously sharing was effected by means of duty cycle. Energy metering devices have long-life lithium batteries, which can operate over very long periods by transmitting at very low duty cycles. The use of LBT would put additional strain on the batteries and significantly reduce their life. At the same time it would be extremely difficult for other LBT devices to detect metering applications where the transmitted telegram would be 1 msec to 40 msec in length and the probability of listening is less than 1/100 whereas the transmit time can be more than a second for LBT devices. This would lead to severe blocking in communication.

- The present SRD sub-bands are becoming more congested and with FHSS and DSSS devices, the noise floor is becoming higher. It is foreseen that there will be insufficient spectrum available to cater for future growth in this market.

The above factors all indicate that the present spectrum for metering is not sufficient and the spectrum for alarms is presently too restrictive. Metering and alarms could share spectrum with automotive and other duty-cycle-only applications.

The installation of smoke detectors is known to reduce the number of people killed or injured by smoke and fire. However, recent research has found that the number of casualties is increasing due to the lack of maintenance (e.g. depletion and/or removal of batteries, etc.), which has rendered the alarms ineffective. This trend can be reversed by the provision of a radio interface to enable monitoring and maintenance This will increase reliability and help to maintain the operational readiness of the alarms. If a fire detector interlink fails due to blocked communications it could result in injury and serious damage to property.

Range

Typical transmission ranges from the remote device to the data collector will be 15 meters to 25 meters, passing though 3 to 5 walls.

Density

On average, a typical dwelling will have between 3 meters and 7 meters. Alarm systems in typical residential installations can include around 20 detectors and activators (smoke, motion, silent alarm buttons, alarm indicators).

Typical data transmitted

The data generated by the meters can include a wide variety of parameters, which can be transmitted to the data collectors. For example, but not limited to, the data may include:

- Consumption
- Flow temperature
- Return temperature
- Errors status
- Date and Time
- Transmission type
- Time of next transmission
- ID-Number
- Tariff
- Alarm status
- Volume flow
- Power



Figure A.6: Wireless heat allocator

Time resolution

For metering and alarm systems, monitoring and control will be performed, on average, once every 2 minutes. Statistics will be gathered, on average, once per day. For alarm systems monitoring intervals can be as short as 10 sec for both intrusion and fire detection systems.

Commercial justification

Commercially, Metering and Alarm applications offer:

- High NRE for the metering equipment
- Long term operation
- High reliability
- Rapid response since for intrusion and fire detection systems
- Monitoring intervals can be as short as 10 sec



Figure A.7: Wireless metering device

In contrast to the other applications in the SRD spectrum, the requirements for alarm systems can be summarized as follows:

- legislation and regulations concerning response time and system life time exist;
- simple and flexible installation of the system, which makes battery operation necessary;
- small ratio of overhead messages in order to extend battery life.

A.3.1.2 Portable alarm details

People working in high-risk environments (such as health care institutions, detention centres or businesses with large compounds) regularly face potential dangers. Rules issued by insurance companies often require tight monitoring of these people, since delay in providing assistance in cases of unforeseen incidents or accidents can be a threat to health and life. Carrying a "personal security" device provides extra safety by means of a manual or automatic alarm to detect a life threatening situation without user intervention, e.g. "no-move", "man-down" or "tear-off".

No dedicated frequency band has yet been designated for ***portable*** alarms. Until now several *non-harmonized* VHF- and UHF-frequencies of the legacy on-site paging standard have to be used. This has the advantage of being interference-free, but has the big disadvantage that for each site in each country a customer-specific frequency should be arranged, together with the corresponding site license.

Goal: to give fast supportive action in emergency by:

- offering help during daily work and emergencies when a person presses the alarm button on his portable alarm device. (manual alarm event; contains also location of person in alarm);
- generating an alarm when the user of a portable alarm device has a "no-move", "tear-off" or "man-down" situation. (automatic alarm event; contains also location of person in alarm);
- location tracking of a person in an alarm situation.



Figure A.8: Portable alarm equipment

System Features:

- Personal Security Management system with downlink to portables.
- 10 to over 1 000 portables with low-power uplink to one or more central receivers (Typically Category 1 as specified in EN 300 220 [i.8]).
- Uplink range is limited due to battery powered transmitter with (low efficiency-) internal antenna, and human body effect- plus multi-wall attenuation of building; typical range is 100 meters to 250 meters.
- System should cover from small (e.g. laboratory) to large areas (e.g. prison, hospital), but always be restricted to "on-site".
- An alarm event should reach the central (host or control unit) within 2 sec.
- The portable should be monitored at least every 10 min.
- Short uplink telegrams comprising "events", to be acknowledged by the central control unit.

Relevant regulations:

- See EC Directives 98/34/EG [i.36] and 98/48/EG [i.37].
- Residential Security Markets [i.6].

Below are some application examples:



Lone workers



Industry



Prison guards



Health care

Figure A.9: Portable alarm applications

A.3.2 Automotive Applications

Automotive applications cover the following functions:

- RKE - Remote Keyless Entry (with additional functions)
- TPMS - Tyre Pressure Monitoring System
- Communication between truck and trailer
- Keyless/Passive Entry
- Passive Start/Stop functions
- Fuel Fired Heater control for trucks and passenger vehicles
- Vehicle Alarm Systems
- Car Communicator

Typical Automotive SRDs transmit bursts of 10 msec to ensure a fast response time. Therefore the LBT technique does not provide acceptable functionality. Furthermore these applications cannot coexist with LBT applications on the same channel.

The applications for SRDs cover comfort and security as well as safety related features:

- **Remote Keyless Entry (RKE)** allows the driver to manually open the vehicle with a radio controlled key. Although representing a feature that mainly increases convenience, they are more and more considered as an essential requirement by customers. Even more convenient is the so-called:
- **Keyless or Passive Entry**, which is based on bidirectional communication between the vehicle and the driver. Proximity sensors built into the door handles detect the approaching driver. A signal is sent from the vehicle to the driver's key, which responds with an identification signal. The same identification signal allows the driver to start the engine by simply pushing a button without putting the key into the ignition lock.
- **Passive Start.** Starting the engine without this identification signal is impossible, which enhances the antitheft protection.
  Additional functions can be integrated into cars as follows:
- **Remote Key**, for opening and closing of windows, sunroof, retractable hardtops etc., doors may automatically lock when the driver exceeds a certain distance from the vehicle. The so-called:
- **Personal Car Communication** allows the driver to receive information at a greater distance from the vehicle telling him e.g. that the lights or radio are left on or that the car is not locked. In addition it may be used to start the Fuel Fired Heater or other car functions. An important aspect is the enhanced safety function of this application e.g. detecting persons left in a car.

- **TPMS (Tyre Pressure Monitoring System)** is considered an integral part of additional measures as required by the European Commission in order to achieve the EU CO2 policy target of 120 g/km.

  Background:

  The EU has set out a strategy to achieve a fleet average target for new vehicles sold in 2012 to 120 g CO_2/km. To achieve this target, the Commission is proposing to require mandatory CO_2 emission targets for manufacturers that equate to average CO_2 emissions of 130 g/km. To compensate for the gap of 10 g/km, additional measures have still to be defined. Current proposals include low rolling resistance tyres, low friction oils, more efficient air-conditioners, TPMS, etc.

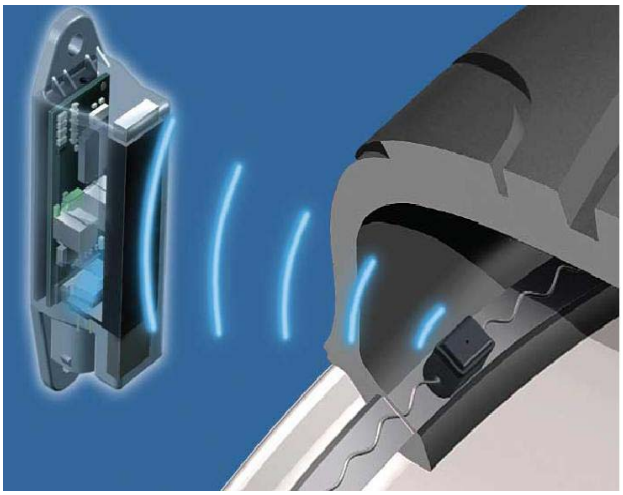
  The role of TPMS is to eliminate the statistically proven incidents of low tyre pressure that lead to increased rolling resistance and consequent higher fuel consumption.

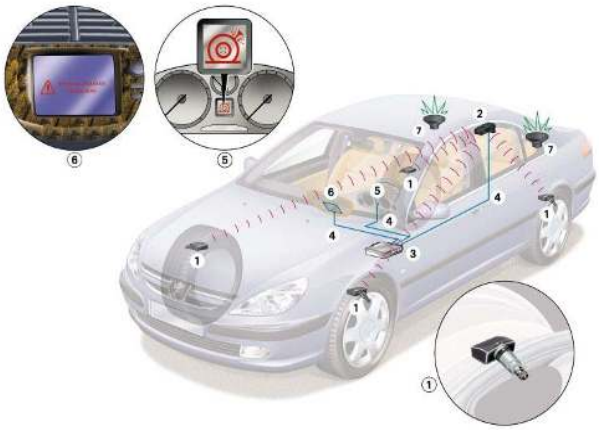
  The safety aspects of TPMS should also not be overlooked. An unreliable system may lead either to failure to warn of low tyre pressures or to false alarms.

- **Communication between truck and trailer** is also provided by SRDs. The transmitted information enables the driver to monitor the status of important points on the trailer, such as brakes, lighting, individual pressure of all tyres etc.

- **Vehicle Alarm Systems**, like "Panic alarm" or "Taxi alarm", are systems which initiate alarm functions like the siren and flashing lights.

Typical illustrations of implemented automotive functions are shown below.





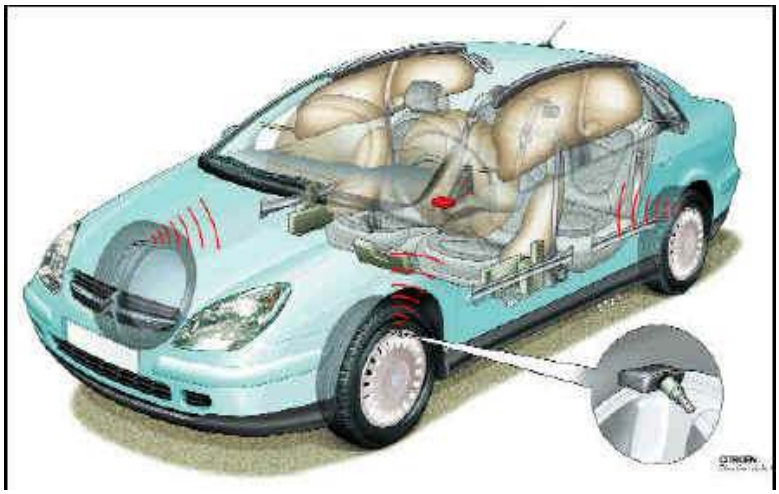


Figure A.10: TPMS Function and implementation



Figure A.11: Car Communicator devices

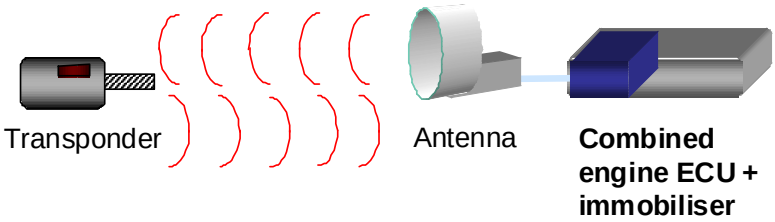
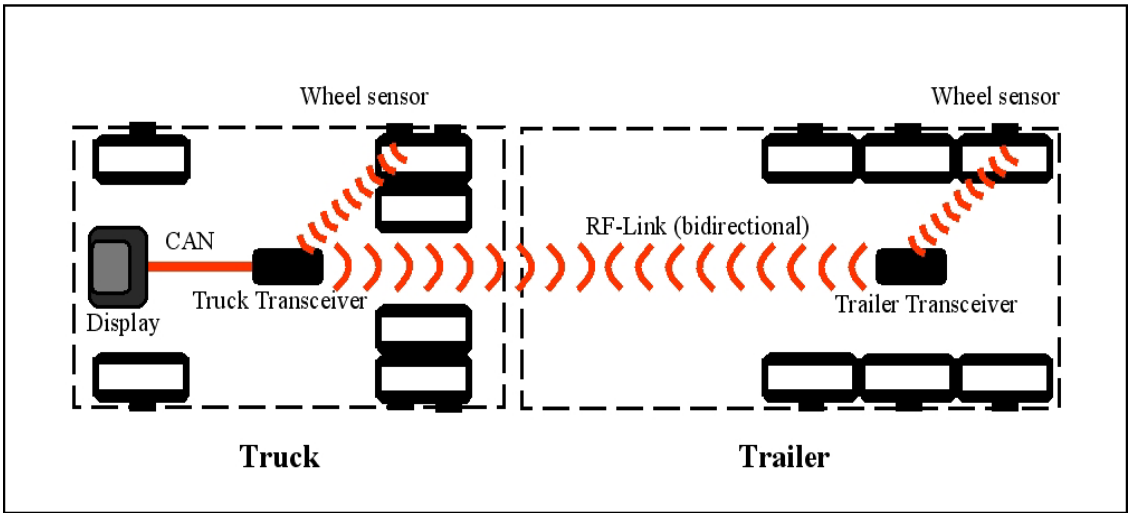


Figure A.12: Combined engine ECU + immobiliser



Figure A.13: Remote Key variants



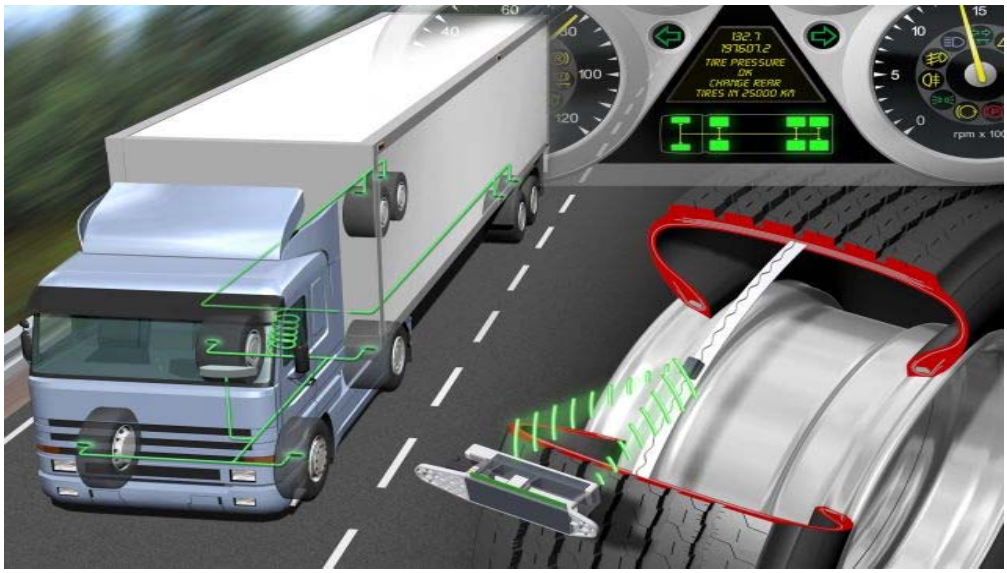


Figure A.14: Truck communicator with TPMS function

Annex B:
Technical information

B.1 RFID

B.1.1 Performance requirements from leading RFID manufacturers and users

The main market requirements are noted in clause A.1.1.

Tags that are used globally are manufactured with their centre frequencies tuned to around 915 MHz. Since tags have a fractional bandwidth of approximately 5 % to 8 %, there will be a noticeable performance penalty if such tags are read by interrogators complying with the present European standard EN 302 208 [i.7]. This is due to the frequency off-set of approximately 45 MHz and the higher transmit powers permitted in other non-European countries.

For comparable performance in Europe, interrogators should operate at around 915 MHz at power levels up to 4 W e.r.p.

B.1.2 Power

By comparison with the limit of 2 W e.r.p. in the current frequency range of 865 MHz to 868 MHz, an increase in the limit to 4 W e.r.p. in the new proposed frequency range of 915 MHz to 921 MHz would lead to the following:

- The read range in free space increases by a factor of 1,4.
- The power absorbed by tags in a pallet is doubled for a given range.
- If the reading performance for tagged items on a pallet reaches 70 % at 2 W, the reading performance increases to 100 % at 4 W. Thus the read probability can be increased from 70 % at 2 W up to 100 % at 4 W (see figure B.1).
- Transponders, which should be aligned parallel to the antenna at their maximum reading range, can be mis-aligned in the same position by up to 60° and still be read.

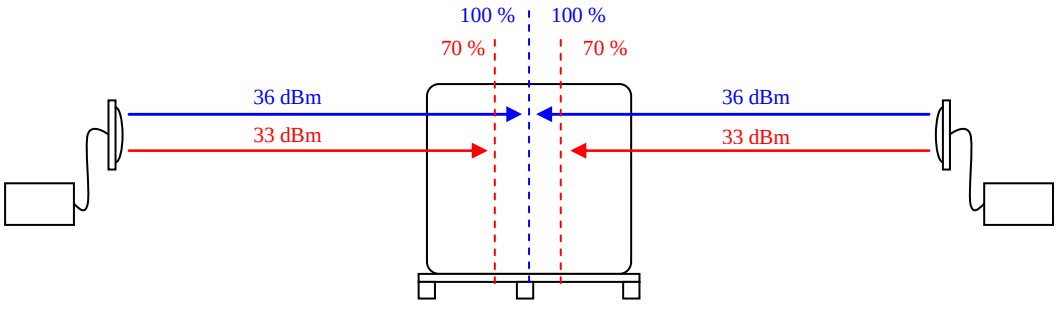


Figure B.1: Comparison of the read range within a pallet

B.1.3 Frequency

The current performance of UHF RFID in the logistic and supply chain market in Europe is limited by the channel bandwidth of only 200 kHz. This restricts UHF RFID in Europe to a reading rate of about 200 tags per second versus a need for 500 tags to 1 500 tags per second for future applications.

To achieve the large-scale roll-out of RFID in Europe, the performance has to fulfil the requirements of all end users. Many end users in the logistic and supply chain market require item level tagging or tagging of small sized cases.

Typically this could mean that there might be 1 500 tagged objects on a single pallet. In order to read these objects on a conveyor belt or in dock door scenario, a higher data rate is required combined with greater penetration of goods by the energizing field.

To achieve an acceptable reading performance the proposed spectrum values are based on the highest data rate in ISO18000-6 [i.10]. In addition to achieve the highest Reader to Tag data rate specified in ISO18000-6C [i.11], a new European UHF RFID band is proposed with the following spectrum parameters:

- The bandwidth for each high power channel should be ~ 400 KHz wide to allow a Tari of 6,25 µs as specified in ISO18000-6C [i.11].
- The channel for the tag response should be 0,8 MHz on both sides of each high power channel. This allows a return link frequency of 640 KHz, which is equivalent to a data rate of 320 kbps.

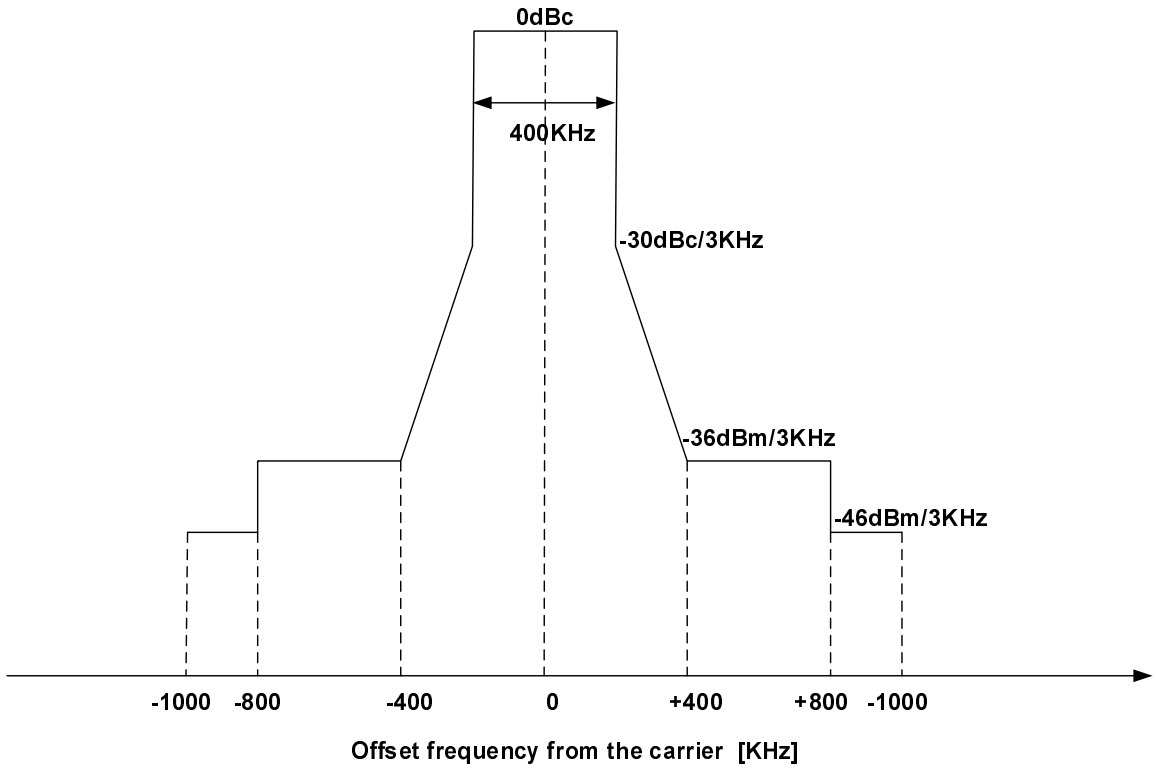


Figure B.2: High performance interrogator channel mask

Receiver parameters:

1) The beamwidth of the antenna(e) in the horizontal orientation should be ≤ 70 degrees.
2) The interrogator should identify a tag with a power level of -65 dBm.
3) The interrogator should identify a tag with a power level of -62 dBm in the presence of an un-modulated blocker with a power level of -35 dBm at a frequency that lies +2 MHz or -2 MHz away from the carrier frequency of the interrogator.
4) The receiver of the interrogator should have a bandwidth of (fc ± 1 MHz).

B.2 Non-specific SRDs

No change. Characteristics of transmitters and receivers should be as defined in EN 300 220 [i.8].

B.3 Specific SRDs

B.3.1 Metering and Alarms

Power

The maximum permitted radiated power should be 25 mW e.r.p.

This will permit a range of 15 meters to 25 meters, with propagation through 3 to 5 walls and a receiver sensitivity of -90 dBm.

Frequency

A data rate of 100 kbit/s is required.

Using FSK, an occupied bandwidth of 150 kHz can be achieved.

Because heat cost allocators are attached to hot water radiators, there is a large operating temperature range and therefore a significant frequency drift. This leads to a required channel width of 250 kHz.

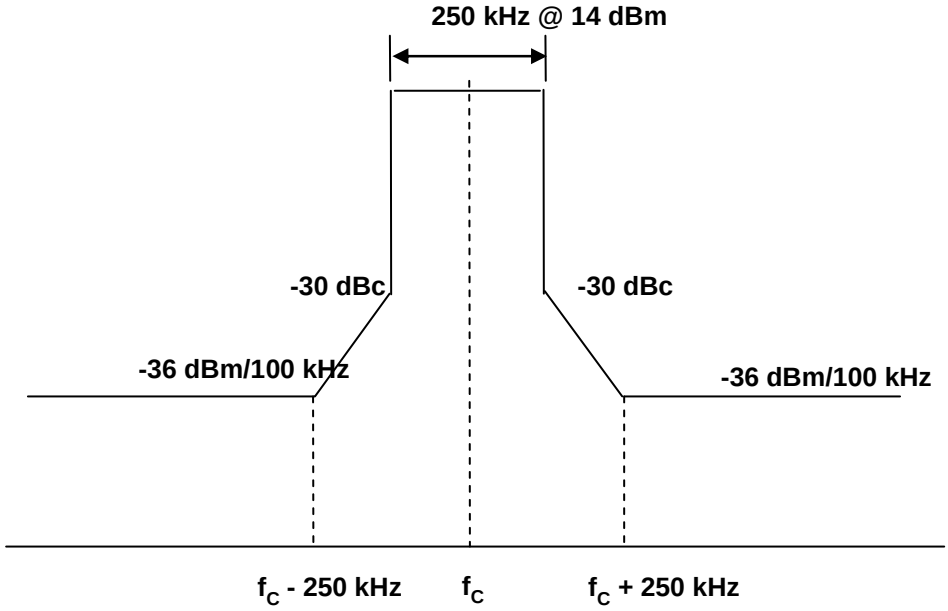


Figure B.3: Metering and Alarms Spectrum Mask

Mitigation techniques

A number of factors provide inherent interference mitigation. The data collectors are indoor applications, so some wall attenuation occurs. The distance from other applications and the very low duty cycles used provide further mitigation.

Duty Cycle

The remote sensors have a duty cycle of $\leq 0{,}1$ %, but the data collectors need to concentrate the data from many sensors, so they need more transmit time. Data collectors typically transmit for about 30 msec with a cycle time of around 30 sec.

The duty cycle should therefore be ≤ 1 %.

Alarm systems that are transmitter-receivers pose problems because energy is wasted due to collisions between messages. In addition, with a duty cycle regulation of 1 % as it is defined today on an hourly basis, it is possible to transmit for as much as 36 sec without interruption. In the LBT enabled band, we currently also see a maximum allowed burst length of up to 4 sec. This makes the use of certain channels impossible for alarm systems, due to the legal requirements for the confirmation of a successful transmission of an alarm message within 10 sec. AFA does not solve the blocked channel problem, because the operation of switching to an unblocked channel takes more time and/or energy than is acceptable.

The solution for robust alarm transmission over radio links is to restrict the channel usage not only to a very low duty cycle possibly with LBT protection, but also to specify an upper bound on transmission burst length (ON-time) and a minimum inter-packet time (OFF-time) allowed on a channel.

B.3.2 Portable Alarms

LBT is not an acceptable technique for "Portable alarms for personal security". Instead they use very low duty cycle access techniques.

The spectrum mask conforms to figure B.4.

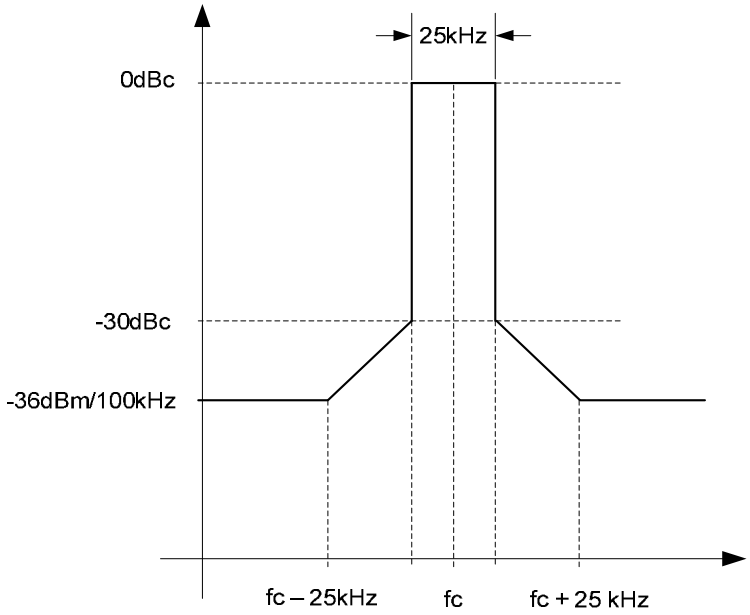


Figure B.4: Portable Alarms Spectrum Mask

- Power:
  The maximum permitted radiated power should be 100 mW e.r.p. This will permit a range of 100 meters to 250 meters in harsh indoor environments (e.g. prisons with reinforced concrete walls and steel doors, industry). A long range is desirable to minimize the number of (very expensive-) central receivers (Typically Category 1 according to
  EN 300 220 [i.8]).

- Duty cycle:
  The portable alarms transmit typically short data telegrams (~250 msec) with a low duty cycle (0,1 %).

- Modulation:
  Narrowband FSK, 25 kHz channel BW.

B.3.3 Automotive

B.3.3.1 Power

The maximum permitted radiated power should be 100 mW e.r.p.

This allows a higher power for transmitting from "car to key" in order to compensate for the low antenna gain of the key due to the reduced size and dimensions of the key-housing. Additional functions like Fuel Fired Heating Systems or Personal Car Communication, needs ranges up to 1 km.

B.3.3.2 Duty Cycle

See table 3 frequency Band g6.

NOTE 1: It should be noted that most automotive systems should respond immediately.

NOTE 2: An example is a passive keyless entry system. For example consider a situation in which a car owner wants to enter his locked car. The door lock device is initiated the moment that the owner touches the door-handle. The system performs a bi-directional communication with the key and, after a successful exchange (e.g. < 80 msec), the mechanical lock will unlock the door (e.g. < 70 msec). The system should complete the whole operation within < 150 msec (Note that a suitable time limit for the user is < 100 msec). If the system takes longer, the user may interpret this as a malfunction.

NOTE 3: Due to the requirement in automotive applications for rapid response times, it is only possible to transmit one or two telegrams. Other techniques like LBT + AFA are not feasible because times to acquire a free channel are too long. In addition further time should be added for synchronizing the key to the car.

B.3.3.3 Frequency

Occupied bandwidth of the transmitted signal

Based on a ASK Modulation with 10 kHz square wave, the required occupied bandwidth is around 300 kHz.

Systems should have sufficient operational reliability. To achieve this, the bandwidth of the designated frequency band should be sufficiently wide.

The frequency band for [ASK] automotive applications should have a bandwidth of at least 1,5 MHz.

As background information consider the following example:

Occupied bandwidth of transmitter:	*300 kHz*
Add additional tolerances due to temperature, aging, initial, XTO pulling,(±100ppm)	*183 kHz*
Total bandwidth	*483 kHz*

It can be assumed that most systems will operate in the centre of their designated frequency band. To avoid a collision it should be possible for systems to provide redundancy and change frequency. The minimum necessary bandwidth for the designated frequency band should be at least 3 × 483 kHz = 1,449 MHz

There are several factors which lead to the required bandwidth of 500 kHz despite the relatively low data rates. *(see also Receiver bandwidth comments).*

The values discussed above under *"background information"* so far (~300 kHz to 500 kHz) are true, if OBW99 % is considered. **These relatively high OBW99 % figures (compared to the used data rates) are caused because of:**

a) Lack of shaping: only few chipsets are available, that offer such techniques; for category 1 and 2 transmitters shaping usually cannot be applied. See explanation below.

b) High FSK deviation: modulation index >> 1, e.g. 10 kbit/s, deviation ±40 kHz).

This is due to HW techniques like crystal pulling, which are not suitable to produce precise deviation over temperature and production spread.

Furthermore, commonly used receiver technologies (limiter based, FM discriminator) benefit from larger deviation in terms of sensitivity and interference robustness.

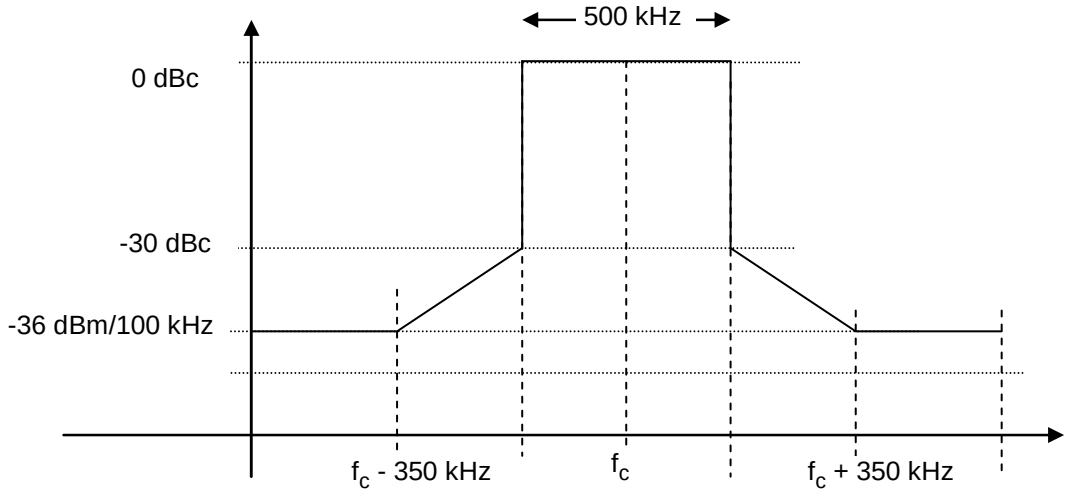
c) Phase noise etc.: caused due to low-power requirements.

This facts makes it difficult to comply with "narrowband-like" ACP (Adjacent Channel Power) or spectrum mask requirements.

In addition, the center frequency tolerance needs to be considered because:

- SAW transmitter: ca. ±200ppm.
- Crystal based PLL transmitters: typically ±100 ppm
  up to ±150 ppm in harsh environments (TPMS, temp. range -40 °C to 125 °C).

B.3.3.4 Spectrum mask



NOTE: Many SRDs in automotive applications have transmitters that use PLLs. Due to the low power levels at which these devices should operate, they generate significant amounts of phase noise. For this reason it is necessary to specify relatively poor limits for the spectrum mask.

Figure B.5: Spectrum mask for automotive

B.3.3.5 Receiver bandwidth

The used transmitter technologies determine the receiver bandwidth.

Three different types of transmitter technology are feasible, which cover automotive designs which are currently used:

1) SAW-oscillator based;
2) crystal based, using either overtone or simple (fixed integer) PLL techniques;
3) sophisticated synthesizer designs, crystal based, using fractional-N PLL techniques and/or shaping techniques.

Types 1 and 2 have the advantage of very low power consumption and cost. Also they meet stringent mechanical and electrical system design requirements.

For certain TPMS solutions SAW-based designs may be required to fulfil robustness requirements from e.g. -40 ºC to +125 ºC temperature, shock and others.

For Type 3 there are only a few IC solutions available, which fulfil automotive qualification requirements, therefore Type 1 and 2 need to be reflected in the receiver bandwidth.

Considering receiver designs, it is sufficient, if the (IF) filter bandwidth covers the main lobe of the transmitted signal, which is ~90 % of the energy.

This "OBW "0 %" bandwidth is significantly smaller (for ASK approx. 2 times the baud rate).

However, the IF bandwidth of the receiver has also to cover the centre frequency tolerance of the transmitter as well as the receiver. Furthermore, widely used IF filter techniques like ceramic filters and integrated analog type filters exhibit considerable variation in filter bandwidth and center frequency, which should also be taken into account.

Thus, typical IF filter bandwidths end up at similar numbers like the OBW99 % (300 kHz to 600 kHz).

B.3.4 Summarizing

The new frequency band should offer enough spectrum to provide for multi-channel systems since under some scenarios one system can use 2 or 3 channels. Instead of using adjacent channels, multi-channel operation should use channels that are spaced well apart. This should ensure that if one channel is subject to interference then the other channel should be free from interference.

The frequency band should provide sufficient channels to allow the operation of a number of systems in a given location. Therefore a spectrum of approximately 3 MHz would allow multi-channel operation as described above.

Annex C:
Expected compatibility issues

C.1 Existing allocations

The European table of allocations for the UHF frequency range of concern is as follows.

870 MHz to 876 MHz paired with 915 MHz to 921 MHz	MOBILE RADIOLOCATION	Defence systems PMR/PAMR	There are no PMR/PAMR system implementations in this frequency band.

Figure C.1: Excerpt of European table of allocations for the frequency ranges under discussion

C.2 Coexistence and sharing issues

The following studies are considered necessary:

- Coexistence studies between the proposed UHF SRD applications in the frequency band 870 MHz to 876 MHz and the GSM-R Mobile Service at 876 MHz (both directions).
- Coexistence studies between the proposed UHF RFID applications in the frequency band 915 MHz to 921 MHz and GSM (MS Tx, BS Rx) at 915 MHz and GSM-R (MS Rx, BS-Tx) at 921 MHz (both directions).
- Sharing studies should be performed to investigate if there are any problems of incompatibility between RFID, SRDs and other users of the band.

Due to the great differences in the usage scenarios of the various radio applications, it is proposed to use the CEPT SEAMCAT simulation tool as the basis for the analysis.

The results from the study should also provide the necessary minimum receiver performance needed by RFID and SRD applications for sharing with the mobile Service in the adjacent band(s).

It is considered that the proposals in the present document can only be implemented on the basis of no protection for SRD/RFID devices operating in the proposed new bands.

Studies may be needed to assess the compatibility between Metering, Alarms, Automotive applications and non-specific SRDs proposed for the band 870 MHz to 876 MHz. The compatibility between all these devices and the allocated services or applications operating in the band above 876 MHz should also be studied.

The conditions for use of the proposed high performance SRD applications, including RFID, in the band 915 MHz to 921 MHz, especially with regard to GSM BS receivers in the band below as well as to GSM-R MS receivers in the band above, will need to be investigated.

RFID and other high power SRD applications may suffer interference from the services in the adjacent bands (GSM MS Tx below 915 MHz and GSM-R BS Tx above 921 MHz) if used in close proximity. The proposed sharing studies should also investigate acceptable conditions and minimum separation distances in combination with appropriate receiver parameters for RFID and other high power SRDs.

Existing ERC/ECC Reports [i.22], [i.23], [i.24], [i.25], [i.26], [i.27], [i.28], [i.29] contain technical information that should be used for conducting the necessary studies.

Annex D:
Letter from TC TETRA MC

TETRA31 (08) 21

European Telecommunications Standards Institute
TC TETRA#31
5-7th March 2008
Sophia Antipolis

To: **ERM**

From: **TC TETRA Management Committee (MC)**

Title: **Liaison Statement (LS) Regarding Use of the 870-876/915-921 MHz Frequency Band**

Date: **25th March 2008**

Dear Gabrielle,

We have no comments to the Draft SRdoc TR 102 649-2 RFID and SRDs in 870 MHz to 876 MHz and 915 MHz to 921 MHz as invited in your 14th March 2008 e-mail. However, we discussed this within the TC TETRA MC and we agreed it might be useful to provide an input to ERM explaining why we believe TETRA has not been deployed in this frequency band since it was first made available to TETRA in ERC Dec (96) 04.

The explanation as to why TETRA has not been deployed in this band falls into four main categories being:

- PMR Market
- Market Forces
- RF Propagation Characteristics
- Technical

PMR Market

The market for TETRA is primarily a replacement market for PMR user organizations that already use PMR networks operating in traditional PMR frequency bands, for example, VHF, 410 MHz to 430 MHz and 450 MHz to 470 MHz, with the exception of European Public Safety and Security (PSS) organizations that have allocated 2 x 5 MHz in the 380 MHz to 400 MHz band. Because of this, existing PMR users want to retain existing base station sites and utilize the same frequency bands to maintain the same RF coverage. As a consequence, the 870 MHz to 876 MHz /915 MHz to 921 MHz band is not required. Also, there are very few, if any, new PMR users that require medium to high capacity PMR networks, being the main requirement met by TETRA.

Market Forces

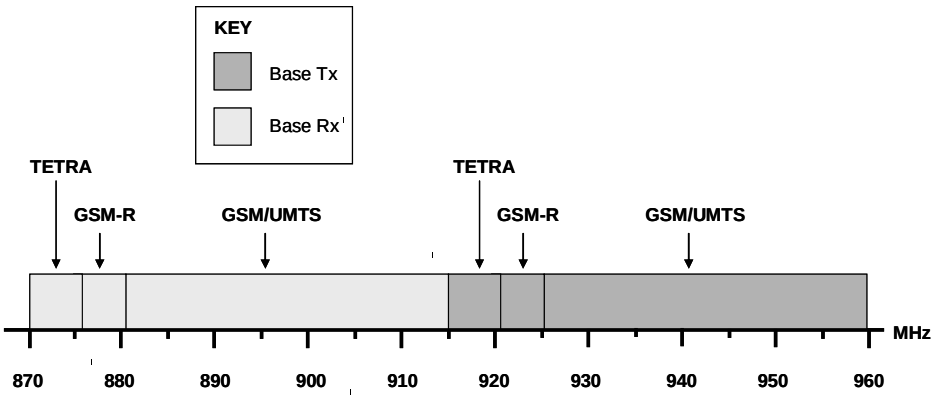
The market represented by the 870 MHz to 876 MHz/915 MHz to 921 MHz Frequency Band resides in Europe only. Also, because the useable spectrum is limited owing to interference (see technical section below) the overall market, compared with the globally harmonized bands of 410 MHz to 430 MHz and 450 MHz to 470 MHz, combined with the 2 x 5 MHz allocated in the 380 MHz to 400 MHz band for European PSS organizations (largest market for TETRA to date), makes the market less attractive, hence the reason why no products have been deployed.

RF Propagation Characteristics

As the majority of TETRA networks utilize multiple base station sites to provide wide area coverage, using the 870 MHz to 876 MHz/915 MHz to 921 MHz band, instead of the 410 MHz to 430 MHz or the 450 MHz to 470 MHz band, would require a greater number of base station sites to provide the same RF coverage. This would be economically unacceptable, especially as most PMR user organizations own their PMR networks.

Technical

An important performance aspect of Mobile Stations (MSs) operating in any duplex band is their receiver selectivity. The chart below explains this in more detail.



As can be seen from the chart the spectrum directly above and below the 915 MHz to 921 MHz (BS to MS) band is used by wideband GSM-R and GSM/UMTS respectively and above the 870 MHz to 876 MHz (MS to BS) band by wideband GSM-R. As a consequence, the guard band required between narrow band TETRA and these wideband technologies reduces the amount of useable spectrum within the 876 MHz/915 MHz to 921 MHz band for TETRA *(see note)*, some estimates indicate as little as 2 x 3 MHz, especially as many traditional PMR user organizations require a high QoS and therefore cannot accept reduced RF coverage or loss of communications caused by interference.

Furthermore, Direct Mode Operation (DMO) in the 870 MHz to 876 MHz/915 MHz to 921 MHz band may also raise complex technical co-existence requirements.

Please note that the content of this LS is not meant to create any formal exchange between ERM and TC TETRA regarding the future use of the 870 MHz to 876 MHz/915 MHz to 921 MHz band but to simply explain why we believe TETRA has not been deployed in this band.

NOTE: The frequency range should read "870 MHz to 876 MHz/915 MHz to 921 MHz" instead of "876 MHz/915 MHz to 921 MHz".

Sincere Regards

Doug Gray

Chairman TC TETRA

History

Document history		
V1.1.1	September 2008	Publication